Does Increasing Government Spending on Tertiary Education Improve the Future Tax Take? A Bayesian Causal Inference Analysis

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Abstract

When a national government increases public spending on tertiary (university) education, does that investment strengthen the country's future tax take once general economic growth is separated out? Cross-country data make this question difficult to answer because faster-growing economies naturally collect more tax revenue, international indicators divide per-student spending by gross domestic product (GDP) per person and student headcount, and university cohorts take several years to graduate and enter higher tax brackets. We assemble a harmonised five-year panel covering 121 countries from 1985 to 2024 and fit Multivariate Bayesian Causal Forests that model four interrelated tax outcomes simultaneously whilst adjusting for starting country conditions and follow-up macroeconomic growth. We find that above-median per-student tertiary funding shifts the tax mix toward direct income, profit, and capital gains taxation by +1.24 percentage points (pp; $P = 0.99$, where P is the posterior probability of an effect greater than zero given the data) after 5 years and +1.48 pp ($P = 0.97$) after 10 years. Overall tax revenue as a share of GDP takes a full decade to respond as university cohorts graduate and advance into higher-salaried careers, with 10-year direct income tax revenue rising by +0.34 pp of GDP ($P = 0.95$) under the baseline spending ratio and by +0.81 pp of GDP ($P = 1.00$) following substantial within-country increases in funding per student. By contrast, expanding the overall university budget without protecting spending per student yields weak fiscal returns.

Keywords: Multivariate Bayesian Causal Forests; tertiary education spending; tax capacity; direct income taxation; cross-country panel data.

1 Introduction

Public debates over higher education finance frequently centre on a simple, practical question: is government spending on tertiary (university) education a net drain on the public purse, or is it a productive investment that expands a country's future tax base? At the individual level, decades of labour economics research have shown that university graduates earn higher wages, experience lower unemployment, and work in larger, more formalised firms than non-graduates (Carneiro et al., 2011; Zimmerman, 2014; Psacharopoulos and Patrinos, 2018; Deming, 2022). Empirical welfare analyses in the United States also indicate that direct investments in college access and per-student instructional resources can recoup a substantial share of their upfront public cost over the lifecycle through higher future income tax payments (Trostel, 2010; Hendren and Sprung-Keyser, 2020; Chakrabarti et al., 2020). As university-educated cohorts enter the labour market and raise firm productivity (Moretti, 2004; Valero and Van Reenen, 2019), those private wage gains and corporate spillovers should in principle show up at the national level as stronger personal income, payroll, and corporate profit tax receipts.

Yet working out whether, and over what time horizon, national government spending on tertiary education actually improves a country's tax take is surprisingly tricky. First, if a country's tax revenue rises over the decade following an increase in university funding, we need to know whether universities specifically

strengthened the formal tax base or whether the whole economy simply grew richer. Worse still, if an analyst constructs a national tax-per-person variable by multiplying the tax-to-GDP ratio by GDP per person, any positive effect of education on GDP automatically inflates the tax outcome by mathematical identity; isolating a genuine fiscal return requires modelling pure tax shares and tax-to-GDP ratios whilst controlling for both starting economic conditions and follow-up economic growth (Imai et al., 2010; Acharya et al., 2016; VanderWeele, 2016). Second, the standard international indicator of tertiary funding intensity compiled by the United Nations Educational, Scientific and Cultural Organisation (UNESCO) Institute for Statistics[1] and the World Bank[2] measures public expenditure per university student as a percentage of GDP per person. Because this ratio places both student headcount and GDP per person in the denominator, a low-income country with a tiny, elite university sector (such as Malawi or Ethiopia, enrolling fewer than 2% of young adults) can appear to spend over 150% of GDP per person per student, whereas a high-income country enrolling 60% of its young adults (such as Denmark or South Korea) appears to spend 27%, even though the high-income country spends nearly three times as many real purchasing power parity (PPP) dollars per student (where PPP adjusts US dollars for differences in local price levels across countries).

A third obstacle is that university investments take time to mature and affect different parts of the tax system unequally. A funding increase in a given five-year period immediately pays formal university salaries and supports student cohorts who take three to four years to graduate and another five years to enter peak tax-paying brackets. Furthermore, public finance theory shows that as economies accumulate skilled workers and formal employers, they shift away from indirect consumption and customs taxes toward direct personal income and corporate profit taxes (Gordon and Li, 2009; Besley and Persson, 2014; Kleven et al., 2016; Jensen, 2022). Looking only at a single headline tax-to-GDP number hides this shift in the underlying tax mix. Figure 1 summarises our study design for addressing all three obstacles across 121 countries from 1985 to 2024.

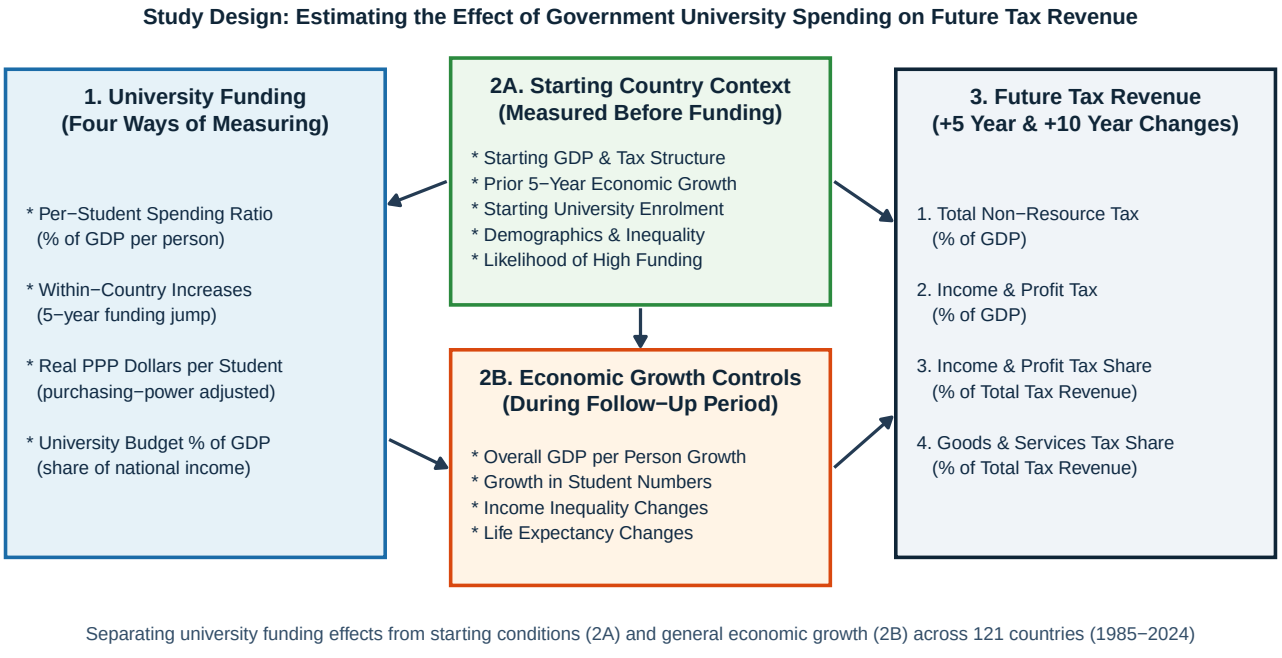


Figure 1: Overview of the study design showing the four ways we measure government university funding, how we control for starting country context (Box 2A) and follow-up economic growth (Box 2B), and the four tax revenue outcomes we track over +5 and +10 years.

As illustrated in Figure 1, we jointly model four percentage-point (pp) tax outcomes over both +5 and +10 years: (1) total tax revenue as a share of GDP (excluding oil and mineral royalties), (2) direct income, profit, and capital gains tax revenue as a share of GDP, (3) direct income and profit taxes as a

[1]UNESCO Institute for Statistics (UIS) education finance database: https://data.uis.unesco.org/.
[2]World Bank World Development Indicators (WDI) portal: https://datatopics.worldbank.org/world-development-indicators/.

share of total tax revenue, and (4) indirect goods and services (sales and value-added) taxes as a share of total tax revenue. Rather than relying on a single composite spending ratio, we compare four observed measures of university funding side by side: the per-student spending ratio (% of GDP per person), within-country per-student funding expansions of at least +2.5 pp over a five-year wave, real 2021 PPP dollars per university student (which removes GDP from the denominator), and the public university budget as a percentage of GDP (which removes student headcount from the denominator). Throughout the paper, we use pp to denote percentage points, CI to denote a Bayesian posterior credible interval, and $P = P(\tau > 0 \mid \text{data})$ to denote the posterior probability that the causal treatment effect τ is greater than zero given the observed data.

We believe this is the first global study to use modern causal inference techniques to test whether public university funding increases future tax revenue. The remainder of the paper is organised as follows. Section 2 reviews the literature on tax capacity, university resources, and Bayesian regression tree models. Section 3 describes the 40-year panel dataset, illustrates the UNESCO denominator trap, and explains where countries such as Ireland sit on the funding scale. Section 4 explains the Multivariate Bayesian Causal Forest (MV-BCF) model in plain language, with full mathematical derivations in Appendix A. Section 5 presents the empirical results across time horizons, funding measures, enrolment levels, and country trajectories, whilst Section 6 discusses the policy implications, limitations, and directions for future research.

2 Literature Review

In this section we review three strands of research that have largely developed in parallel: the political economy of tax structure modernisation, the economics of per-student university resources, and Bayesian tree ensembles for causal inference. Bringing these three literatures together clarifies why university resources should affect direct income taxation more strongly than consumption taxation, and why flexible nonparametric models are needed to separate that fiscal return from general economic growth. Those interested in more comprehensive surveys of each field may refer to Besley and Persson (2014), Deming (2022), and Hill et al. (2020).

2.1 Tax Capacity, Formal Employment, and Tax Structure

A central question in public finance is why high-income countries collect 30% to 45% of GDP in tax revenue (primarily through direct personal income taxes, social security contributions, and corporate profit taxes), whereas low- and middle-income countries often collect less than 15% of GDP and rely heavily on indirect consumption taxes and customs duties (Gordon and Li, 2009; Besley and Persson, 2009, 2014). Early cross-country studies emphasised political institutions, property rights, and administrative state capacity (Besley and Persson, 2009; Kaufmann et al., 2011), alongside improved international accounting standards designed to separate domestic tax effort from volatile oil and mineral royalties (Prichard, 2016; UNU-WIDER, 2023).[3] Whilst these institutional studies establish that fiscal capacity evolves alongside broader economic development, they leave open the specific microeconomic channels through which public investments in skilled human capital expand the collectible tax base.

More recent empirical work has focussed on the microeconomic mechanics of tax enforcement, paper trails, and formal employment (Kleven et al., 2016; Keen and Slemrod, 2017; Pomeranz and Vila-Belda, 2019). Kleven et al. (2016) show that modern governments can enforce high direct tax compliance at low audit cost because large, formal firms act as third-party tax collectors: complex businesses require internal accounting records shared across managers and skilled staff, making collusive tax evasion between employers and employees unsustainable. Building on this mechanism, Jensen (2022) analyses roughly 100 household surveys across development levels and shows that as the share of salaried employees rises relative to self-employed workers, governments lower personal income tax exemption thresholds and expand direct income tax collection. Field studies confirm that electronic tax filing and third-party

[3] See the UNU-WIDER Government Revenue Dataset (GRD) documentation at https://doi.org/10.35188/UNU-WIDER/GRD-2023.

information trails reduce evasion when paired with capable firms and literate taxpayers (Pomeranz and Vila-Belda, 2019; Okunogbe and Pouliquen, 2022), whilst broader direct taxation can also strengthen citizen engagement with the state (Weigel, 2020). By contrast, in developing economies with large informal sectors, indirect consumption taxes such as value-added tax (VAT) remain the primary revenue instrument because formal retail chains only gradually replace informal vendors as incomes rise (Bachas et al., 2024).

This public finance literature suggests a direct connection between university funding and the national tax base that goes beyond headline GDP growth. Universities are themselves large formal employers whose academic and administrative staff pay income tax through automatic payroll withholding, and they train the professional, managerial, scientific, and accounting graduates required by formal corporate employers (Kleven et al., 2016; Valero and Van Reenen, 2019), shifting workers into salaried jobs above national income tax exemption thresholds (Jensen, 2022). When this mechanism operates at the national level, higher public spending per university student should specifically expand direct income, profit, and capital gains taxation as a share of total tax revenue and as a share of GDP, whilst reducing relative reliance on indirect goods and services taxes.

2.2 Returns to University Resources and Fiscal Externalities

A large microeconometric literature documents strong private wage returns to university education across both advanced and developing economies (Carneiro et al., 2011; Zimmerman, 2014; Psacharopoulos and Patrinos, 2018; Chetty et al., 2020; Mountjoy, 2022). Reviewing this evidence, Deming (2022) notes that human capital accounts for at least one-third of within-country wage variation and at least half of cross-country income differences, with cognitive and problem-solving skills playing a key role in long-run growth (Hanushek and Woessmann, 2012; Angrist et al., 2021), even after accounting for skilled emigration from smaller developing countries (Beine et al., 2008). Beyond individual earnings, university graduates generate productivity spillovers for surrounding workers and firms: Moretti (2004) finds that a higher share of college graduates in a city raises the wages of both graduates and non-graduates, whilst Valero and Van Reenen (2019) examine roughly 15,000 universities across 1,500 sub-national regions in 78 countries and find that university expansions raise future regional GDP per person through skilled labour supply and local innovation.

Crucially for our study, higher education research demonstrates that graduation rates and subsequent earnings depend heavily on per-student instructional resources rather than enrolment headcounts alone. Bound et al. (2010) and Deming and Walters (2017) show that when public colleges expand student numbers whilst state funding per student stagnates or declines (a phenomenon known as institutional crowding), degree completion rates fall and time to graduation lengthens. Conversely, Chakrabarti et al. (2020) find that increases in state appropriations to public universities raise bachelor's degree completion and boost graduates' earnings into their thirties. When graduates complete their degrees and enter formal careers, the fiscal return to the public purse can be substantial: Trostel (2010) estimates that lifetime additional income and payroll taxes paid by college graduates exceed direct government subsidies, and Hendren and Sprung-Keyser (2020) show within their Marginal Value of Public Funds framework that direct investments in college access and per-student spending generate positive fiscal externalities through future tax receipts (though they also find that untargeted higher education tax credits for adults yield lower welfare returns than early childhood investments).

2.3 Bayesian Regression Tree Ensembles for Causal Inference

Separating the fiscal effect of university funding from general economic development involves two statistical challenges: controlling for intermediate economic growth without introducing bias, and accounting for non-linear confounding across diverse countries. In cross-country panels, standard linear two-way fixed effects regressions can suffer from negative weighting when treatment effects vary across countries and over time (Jordà, 2005; Acemoglu et al., 2019; de Chaisemartin and D'Haultfœuille, 2020; Callaway and Sant'Anna, 2021). Furthermore, as shown in the causal mediation literature (Imai et al., 2010; VanderWeele, 2016; Acharya et al., 2016; Ting and Linero, 2025), adjusting for follow-up macroeconomic

changes (such as GDP per person growth, enrolment growth, inequality, and life expectancy) allows us to isolate the direct structural effect of university spending net of overall economic growth, especially when checked via two-stage sequential g-estimation (Acharya et al., 2016).

Bayesian Additive Regression Trees (BART) (Chipman et al., 2010; Hill, 2011; Hill et al., 2020) and Bayesian Causal Forests (BCF) (Hahn et al., 2020) provide a flexible nonparametric alternative to linear panel regressions, sitting alongside frequentist causal forests and double machine learning (Chernozhukov et al., 2018; Athey et al., 2019). Tree ensembles automatically capture non-linear relationships and multi-way interactions amongst covariates, with extensions supporting high-dimensional predictors (Linero, 2018), functional smoothing (Starling et al., 2020), count outcomes (Murray, 2021), fast stochastic tree search (Krantsevich et al., 2023), causal mediation (Ting and Linero, 2025), and low-dimensional posterior summaries (Woody et al., 2021). Hahn et al. (2020) show that standard single-forest BART models can suffer from regularisation-induced confounding when strong baseline predictors of the outcome also predict which countries receive higher funding; BCF resolves this by splitting the model into a prognostic forest μ (which includes the estimated propensity score) and a separately regularised treatment-effect forest τ, with extensions to multivariate outcomes and longitudinal panels (McJames et al., 2025, 2024). Readers will note that the Multivariate Bayesian Causal Forest (MV-BCF) method used in this paper (McJames et al., 2025) was co-developed by the author with Nathan McJames, Ann O'Shea, and Yong Kai Goh. Given this, we want to make it clear from the outset that this paper is not written to promote one causal estimator over all others; in Appendix B, we compare the MV-BCF model directly against univariate BCFs, single-forest BART, linear local projections, two-way fixed effects, and double machine learning, and we discuss the limitations of the BCF approach alongside its strengths.

3 Data

3.1 Global Five-Year Panel Structure

Annual macroeconomic series for education spending and tax revenue contain frequent reporting gaps and short-term business-cycle fluctuations. Following standard practice in empirical macro-panel studies (Acemoglu et al., 2019), we average annual observations from the World Bank WDI, the UNESCO Institute for Statistics, the International Monetary Fund (IMF) Fiscal Monitor,[4] the Worldwide Governance Indicators (WGI), the Standardised World Income Inequality Database (SWIID version 9.6; Solt 2020),[5] and the Organisation for Economic Co-operation and Development (OECD) *Education at a Glance* financing taxonomy[6] into eight non-overlapping five-year waves spanning 40 years: `1985-1989`, `1990-1994`, `1995-1999`, `2000-2004`, `2005-2009`, `2010-2014`, `2015-2019`, and `2020-2024`. Constructing five-year waves also matches the typical duration of an undergraduate degree programme and reduces serial correlation from annual budgetary carryover.

Although the full World Bank register lists 217 sovereign territories, many small island states and conflict-affected territories do not report tertiary education expenditure or disaggregated tax revenues to UNESCO and the World Bank. In addition, raw international fiscal archives occasionally contain reporting anomalies where currency-unit mismatches or offshore sovereign resource funds distort headline ratios. Screening out impossible tax-to-GDP values above 60% or below 0.5% of GDP removes two raw WDI reporting errors: Sudan in `1995-1999` (recorded at 629.2% of GDP due to a currency-unit mismatch) and Timor-Leste in `2010-2014` (recorded at 110.0% of non-oil GDP due to offshore petroleum fund accounting). Requiring strictly observed university funding and four-outcome tax series (with zero imputation on treatments or outcomes) yields our analysis dataset of 121 countries: 372 five-year country-waves across 121 countries at the +5-year horizon ($t \to t+1$) and 283 country-waves across 112 countries at the +10-year horizon ($t \to t+2$).

[4] IMF Fiscal Monitor database: `https://www.imf.org/en/Publications/FM`.
[5] SWIID version 9.6 database and documentation: `https://fsolt.org/swiid/`.
[6] OECD *Education at a Glance* portal: `https://www.oecd.org/en/publications/education-at-a-glance.html`.

3.2 Tax Revenue Outcomes

To examine how university funding affects both overall tax collection and the mix of taxes without introducing mechanical GDP product identities, we track four percentage-point changes over +5 years ($t+1$) and +10 years ($t+2$). First, we measure total tax revenue as a percentage of GDP (ΔTax/GDP, in pp of GDP), based on the World Bank WDI tax-to-GDP series (`GC.TAX.TOTL.GD.ZS`). Following the fiscal-capacity literature (Prichard, 2016; UNU-WIDER, 2023), in major oil- and mineral-exporting countries where natural resource rents exceed 10% of GDP, we subtract half of the excess resource rents so that volatile commodity royalties do not masquerade as domestic tax effort (and we verify in Appendix B that using raw, unadjusted WDI tax-to-GDP gives virtually identical results).

Second, we measure direct income, profit, and capital gains tax revenue as a percentage of GDP (ΔIncomeTax/GDP, in pp of GDP), which captures revenue collected directly from individual wages, salaries, and corporate profits. Third, we track direct income, profit, and capital gains taxes as a percentage of total revenue (ΔIncomeTax/Rev, in pp of revenue), which measures structural modernisation toward progressive direct taxation (Besley and Persson, 2014; Jensen, 2022). Fourth, we track indirect goods and services taxes (such as VAT, sales taxes, and excise duties) as a percentage of total revenue (ΔGoodsTax/Rev, in pp of revenue; Bachas et al. 2024). Readers should note that Outcome 2 (direct income tax as a share of GDP) is simply the product of Outcome 1 (total tax as a share of GDP) and Outcome 3 (direct income tax as a share of revenue) divided by 100. Thus, Outcomes 1, 3, and 4 capture the two underlying policy dimensions (whether overall tax collection rises and whether the tax mix shifts from consumption taxes toward income taxes), whilst Outcome 2 measures their combined impact on the national budget. Because these four measures are algebraically linked and move together within each country's budget, a major advantage of our Multivariate Bayesian Causal Forest approach is that it models all four outcomes simultaneously through a joint covariance structure, capturing their cross-outcome dependencies directly unlike standard single-equation regressions.

3.3 The UNESCO Denominator Trap and Country Funding Thresholds

The standard international indicator of per-student university spending compiled by UNESCO and the World Bank divides public spending per university student by GDP per person, which is algebraically identical to dividing the public university budget (% of GDP) by the number of university students per person:

$$\text{UNESCO Ratio}_{i,t} = \frac{\text{University Spend per Student}_{i,t}}{\text{GDP per Person}_{i,t}} \times 100 = \frac{\text{University Budget (\% GDP)}_{i,t}}{\text{Students per Person}_{i,t}}. \quad (1)$$

Because both GDP per person and student headcount appear in the denominator, countries with very different university systems can record similar headline ratios for completely different reasons. To see the raw data before any statistical modelling, Figure 2 plots the observed distributions of the UNESCO per-student ratio, real 2021 PPP dollars per student, gross tertiary enrolment, and the direct income tax share across the four World Bank income groups, and Table 1 reports the corresponding group summaries.

Figure 2 and Table 1 show clearly how the UNESCO denominator trap works in practice. In low-income countries (such as Uganda, Malawi, or Ethiopia), only 2.3% of young adults attend university; dividing a small university budget (0.47% of GDP) by a tiny student headcount and a low GDP per person produces a raw median UNESCO ratio of 205% (which we cap at 150% in Figure 2A and when fitting our regression trees so that extreme outliers do not distort the tree splits). That tiny elite student body also explains why median real PPP spending per student in low-income countries ($3,957, concentrated in one or two national universities) is slightly higher than in lower-middle-income countries ($3,258), where enrolment is over three times higher (7.4% median) and public budgets are stretched across many more students. Once we unbundle the denominators in Table 1, high-income countries (such as Denmark, Norway, or South Korea) spend nearly three times as much per student in real purchasing power ($11,148 vs $3,957) and twice as much of their national income (0.94% vs 0.47% of GDP), whilst collecting a larger share of their tax revenue from direct income and profit taxes (38.4% vs 29.2%).

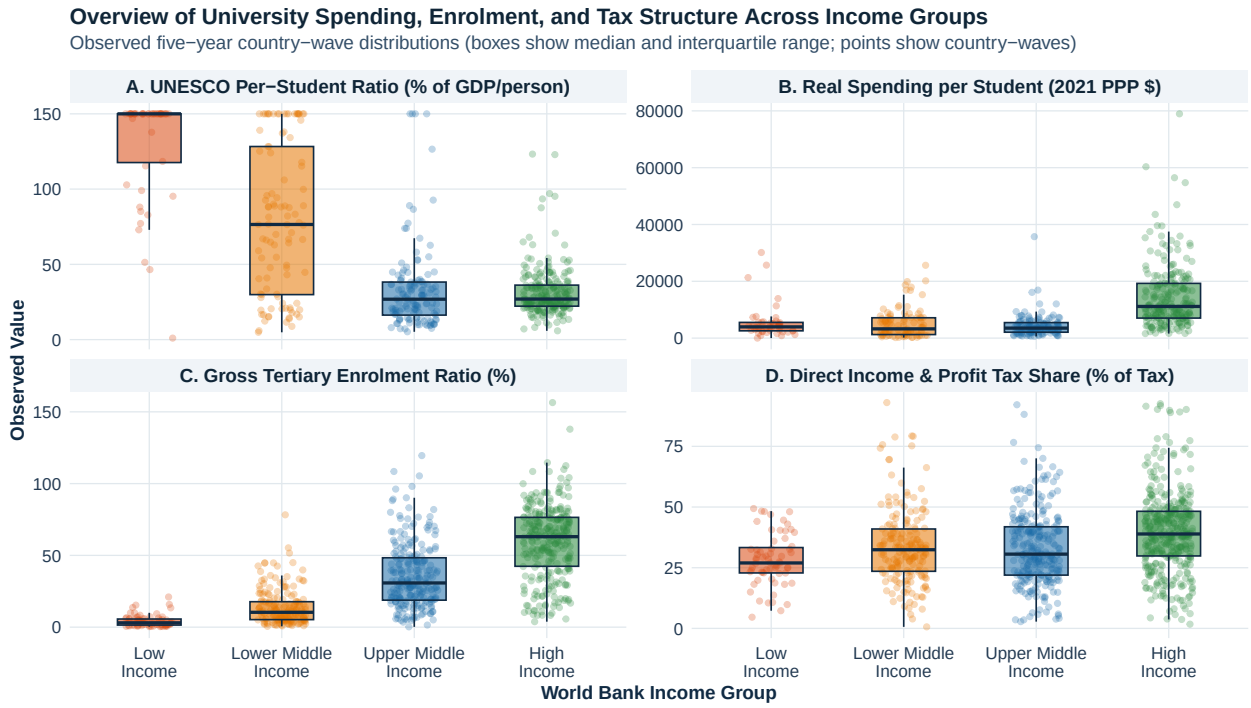


Figure 2: Observed distributions of (A) the UNESCO per-student spending ratio (% of GDP per person, capped at 150% for display), (B) real spending per university student in 2021 PPP dollars, (C) gross tertiary enrolment ratio (%), and (D) direct income and profit tax share (% of revenue) across the four World Bank income groups.

Table 1: University spending indicators, enrolment, and baseline direct income tax share across World Bank income groups. Values in brackets in the first column show the number of countries and the number of observed five-year country-waves; group medians are reported for skewed spending and enrolment variables and group means for the direct income tax share.

Income Group (Countries, Waves)	UNESCO Ratio (%)	Real PPP $ / Student	Univ. Budget (% GDP)	Tertiary Enrol. (%)	Direct Income Tax Share (%)
Low income (21, 80)	**205%**	$3,957	0.47%	2.3%	29.2%
Lower middle (41, 172)	76.5%	$3,258	0.65%	7.4%	30.8%
Upper middle (54, 219)	26.8%	$3,521	0.64%	25.5%	33.1%
High income (69, 337)	26.9%	**$11,148**	**0.94%**	**58.3%**	**38.4%**

Because comparing raw UNESCO ratios across income groups would classify almost every low-income country as high-spending and almost every high-income country as low-spending, we define our primary binary funding indicator ($Z_{i,t} = 1$) by comparing each country's per-student spending ratio against the median of its own World Bank income group (205% for low income, 76.5% for lower-middle income, 26.8% for upper-middle income, and 26.9% for high income). This within-group threshold places roughly half of all country-waves in the above-median funding group (47.3% at +5 years and 49.8% at +10 years). Among high-income countries, Nordic and continental European countries with strong public university investment (such as Norway at 38.6% to 45.8%, Sweden at 43.2% to 54.2%, Denmark at 41.5%, and Austria at 35.0% to 36.2%) sit consistently above the 26.9% threshold ($Z_{i,t} = 1$), whereas countries that rely more heavily on private tuition or have lower public per-student allocations (such as Chile at 19.6%, the United States at 20.9%, and Spain at 22.4%) sit below it ($Z_{i,t} = 0$). Ireland provides a particularly clear example of a country that moved across the threshold over time: Ireland sat just above the high-income median ($Z_{i,t} = 1$) in `1995–1999` (28.2%), `2005–2009` (28.4%), and `2010–2014` (27.6%), but fell below the median ($Z_{i,t} = 0$) in `2000–2004` (25.6%) and dropped sharply to 17.3% in `2015–2019` as student numbers expanded much faster than state university funding after the 2008 financial crisis. In Appendix Table 7, we list all 48 countries from our 10-year modelling sample that sit in the below-median group ($Z_{i,t} = 0$) in the most recent treatment wave (`2015–2019`) and therefore stand to benefit from moving to $Z_{i,t} = 1$, ranging from countries sitting right on the boundary (such as Portugal at 26.9% and Poland at 26.8%) to countries substantially below their income-group median (such as Ireland at 17.3%, South Korea at 14.9%, Colombia at 20.6%, Argentina at 16.7%, and Bangladesh at 29.9%).

4 Methodology

In an ideal experiment, we would randomly assign some countries to increase university funding and others to hold funding steady, and then track their tax receipts over the next decade. In observational data, however, governments set their university budgets in response to economic conditions, demographics, and existing tax capacity, and the relationships among these variables are strongly non-linear (Hill, 2011; Hill et al., 2020). To separate the causal effect of university funding from confounding macroeconomic trends whilst modelling all four interrelated tax outcomes simultaneously, we use the Multivariate Bayesian Causal Forest (MV-BCF) model (Hahn et al., 2020; McJames et al., 2025), providing the intuitive setup here and full mathematical derivations in Appendix A.

The Multivariate Bayesian Causal Forest (Hahn et al., 2020; McJames et al., 2025) splits the prediction of future tax changes into two distinct regression tree ensembles (Chipman et al., 2010): a baseline prognostic trajectory (what would have happened to a country's tax take naturally given its starting conditions and overall economic growth) and a causal treatment effect that can vary with country characteristics. For country i at wave t and follow-up horizon $h \in \{1, 2\}$ (+5 and +10 years), the four tax changes $\Delta Y_{i,t+h} \in \mathbb{R}^4$ are modelled jointly as:

$$\Delta Y_{i,t+h} = \underbrace{\mu(X_{i,t}, \hat{\pi}(X_{i,t}))}_{\text{Baseline Trajectory}} + \underbrace{\tau(W_{i,t})}_{\text{Causal Effect}} \cdot Z_{i,t} + \varepsilon_{i,t+h}, \quad \varepsilon_{i,t+h} \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(0, \Sigma). \tag{2}$$

To estimate the baseline trajectory $\mu(X_{i,t}, \hat{\pi}_{i,t})$, we use an ensemble of 75 shallow decision trees (many having only one or two splits), where each terminal leaf outputs a vector of four values corresponding to our four tax outcomes. The baseline forest is fed with 19 pre-treatment country characteristics measured at or before wave t (starting log real GDP per person in PPP dollars, starting total tax-to-GDP ratio, starting direct income tax share, starting goods and services tax share, disposable income Gini inequality, life expectancy, gross tertiary and secondary enrolment ratios, total education spending as a share of GDP, working-age population share, log population, unemployment rate, natural resource rents, trade openness, lagged five-year GDP per person growth, World Bank income group, geographic region, OECD membership, and wave time index) together with four follow-up macroeconomic controls over $[t, t + h]$ (forward GDP per person growth, university enrolment expansion, income inequality changes, and life expectancy changes) so that general economic growth during the follow-up window is separated out (Imai et al., 2010; Acharya et al., 2016). We also include the pre-treatment propensity

score $\hat{\pi}(X_{i,t}) = P(Z_{i,t} = 1 \mid X_{i,t})$ inside $\mu(\cdot)$ to guard against regularisation-induced confounding (Hahn et al., 2020), and verify in Appendix C that treated and control observations exhibit strong propensity overlap across $[0.041, 0.960]$.

To allow the fiscal return to vary across countries without chasing noise in a macro panel, we model the treatment effect $\tau(W_{i,t})$ using a second, smaller ensemble of 30 shallower trees fed with 8 country moderators $W_{i,t}$ (gross tertiary enrolment, log real GDP per person, gross secondary enrolment, starting direct income tax share, working-age population share, World Bank income group, OECD membership, and wave index). Using a larger, more flexible tree prior for the prognostic forest $\mu(\cdot)$ and a smaller, more conservative tree prior for the treatment-effect forest $\tau(\cdot)$ is standard practice in Bayesian causal forest modelling (following Hahn et al. 2020; see also Chipman et al. 2010 and Hill 2011), as it allows the model to absorb complex baseline confounding whilst shrinking noisy subgroup treatment differences toward the overall mean. All four tax outcomes are linked through the 4×4 residual covariance matrix Σ, which receives a conjugate Inverse-Wishart prior (with all prior hyperparameters detailed in Appendix A).

The main advantage of the MV-BCF framework is that it captures non-linear confounding and shares information across correlated tax outcomes whilst providing full Bayesian posterior distributions for every country and horizon. The main disadvantage is that tree ensembles rely on Markov chain Monte Carlo (MCMC) sampling and conditional ignorability assumptions rather than automatic linear fixed-effect differencing. All MV-BCF models are fitted in C++ for 2,000 MCMC iterations (750 burn-in, 1,250 retained posterior draws), with convergence and posterior predictive fit diagnostics reported in Appendix C.

5 Results

5.1 Effects on Tax Revenue After 5 and 10 Years

We begin by examining our primary question: when a country spends above its income-group median per university student, what happens to its tax revenue 5 and 10 years later once starting country conditions and follow-up economic growth are controlled for? Recall that throughout all tables and figures, CI denotes a Bayesian posterior credible interval and $P = P(\tau > 0 \mid \text{data})$ denotes the posterior probability that the treatment effect is greater than zero given the data. Table 2 and Figure 3 report the posterior mean treatment effects, 80% and 95% credible intervals, and posterior probabilities P across the four tax outcomes at +5 years ($N = 372$, 121 countries) and +10 years ($N = 283$, 112 countries).

Table 2: Posterior mean treatment effects (in percentage points, pp), 80% and 95% credible intervals (CIs), and posterior probabilities $P = P(\tau > 0 \mid \text{data})$ from the four-outcome Multivariate Bayesian Causal Forest at +5 years ($N = 372$, 121 countries) and +10 years ($N = 283$, 112 countries).

Tax Revenue Outcome	+5 Years Mean (P)	+5 Years 80% CI / 95% CI	+10 Years Mean (P)	+10 Years 80% CI / 95% CI
1. Total Non-Resource Tax / GDP (% of GDP, pp)	+0.02 ($P = 0.55$)	$[-0.30, +0.32]$ $[-0.46, +0.49]$	+0.45 ($P = 0.86$)	$[-0.07, +0.98]$ $[-0.35, +1.24]$
2. Direct Income & Profit Tax / GDP (% of GDP, pp)	+0.21 ($P = 0.95$)	$[+0.04, +0.38]$ $[-0.04, +0.47]$	+0.34 ($P = 0.95$)	$[+0.07, +0.63]$ $[-0.06, +0.76]$
3. Direct Income & Profit Tax Share (% of total tax revenue, pp)	+1.24 ($P = 0.99$)	$[+0.57, +1.96]$ $[+0.24, +2.40]$	+1.48 ($P = 0.97$)	$[+0.54, +2.50]$ $[-0.11, +3.03]$
4. Goods & Services Tax Share (% of total tax revenue, pp)	−0.29 ($P = 0.31$)	$[-0.97, +0.39]$ $[-1.41, +0.73]$	−0.56 ($P = 0.24$)	$[-1.58, +0.51]$ $[-2.02, +1.05]$

Table 2 and Figure 3 show a clear pattern across the two time horizons. First, higher per-student university funding produces an immediate and lasting shift in the tax mix toward direct income, profit, and capital gains taxation: the direct income tax share of revenue (Outcome 3) rises by +1.24 pp after 5 years (95% CI $[+0.24, +2.40]$, $P = 0.99$) and by +1.48 pp after 10 years (95% CI $[-0.11, +3.03]$, $P = 0.97$; in Appendix B we note that omitting the follow-up macroeconomic growth controls changes

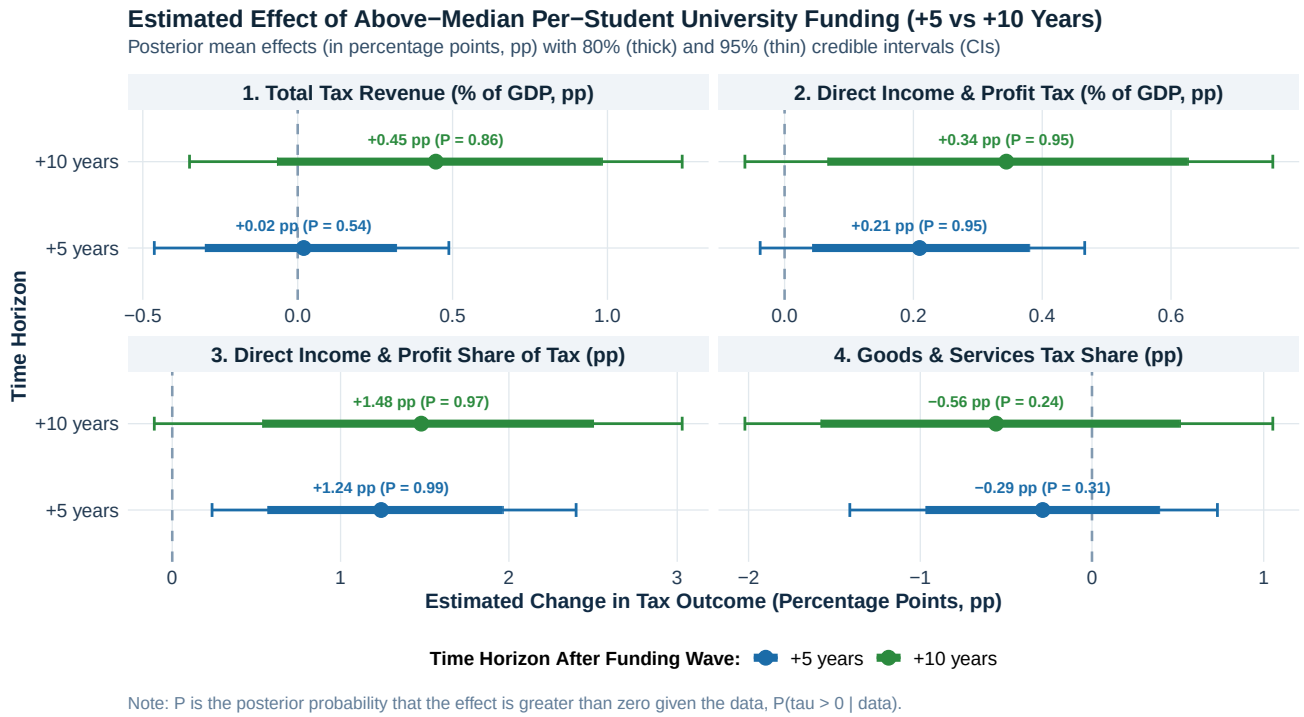


Figure 3: Posterior mean treatment effects (in percentage points, pp) with thick 80% and thin 95% credible intervals (CIs) and posterior probabilities $P = P(\tau > 0 \mid \text{data})$ across the four tax outcomes at +5 years (blue) and +10 years (green).

these estimates by less than 0.25 pp, giving +1.02 pp at +5 years and +1.36 pp at +10 years). Second, overall tax collection as a share of GDP takes a full decade to respond: after 5 years, total non-resource tax revenue (Outcome 1) is essentially unchanged at +0.02 pp of GDP (95% CI $[-0.46, +0.49]$, $P = 0.55$), whereas after 10 years, as university cohorts complete their degrees and advance into higher-salaried careers, total non-resource tax revenue rises by +0.45 pp of GDP (95% CI $[-0.35, +1.24]$, $P = 0.86$) and direct income tax revenue as a share of GDP (Outcome 2) rises from +0.21 pp at +5 years ($P = 0.95$) to +0.34 pp of GDP at +10 years (80% CI $[+0.07, +0.63]$, 95% CI $[-0.06, +0.76]$, $P = 0.95$). Third, on the indirect goods and services tax share (Outcome 4), the estimated effect is slightly negative in share terms (−0.29 pp at +5 years, $P = 0.31$; −0.56 pp at +10 years, $P = 0.24$) with wide credible intervals spanning zero, confirming that the fiscal return is concentrated in direct income and profit taxation rather than consumption taxes. To put these percentage-point figures into concrete national currency terms (holding each country's latest population and 2021 PPP GDP per person constant so that demographic growth does not inflate the calculation), a 10-year gain of +0.34 pp of GDP in direct income tax revenue corresponds to roughly €0.62 billion ($0.68 billion) per year in additional income tax receipts for the median-sized economy in our sample (total GDP of €183 billion), or around €1.2 billion per year in a medium-sized high-income economy such as Ireland and €11.0 billion per year in a large economy such as the United Kingdom (we report percentage points of GDP in cross-country charts because national GDP spans four orders of magnitude across our 121 countries).

5.2 Comparing the Four Ways of Measuring University Funding

Whilst Figure 3 compared the +5-year and +10-year time horizons for our baseline per-student spending ratio, a natural follow-up question is how the +10-year fiscal return changes when we unbundle the UNESCO ratio into its separate components. In Table 3 and Figure 4, we hold the time horizon fixed at +10 years and compare four distinct ways of measuring university funding: (1) our baseline per-student spending ratio (% of GDP per person), (2) observed within-country per-student funding expansions of at least +2.5 pp (and at least +2.5% relative growth) over a five-year wave, (3) real 2021 PPP dollars per student (which removes GDP per person from the denominator), and (4) the total public university budget as a percentage of GDP (which removes student headcount from the denominator). Comparing

these four definitions on the same 10-year horizon allows us to test whether future tax gains are driven by real instructional resources per student or merely by aggregate budget size.

Table 3: Comparison of +10-year posterior mean effects (in percentage points, pp), 95% credible intervals (CIs), and posterior probabilities $P = P(\tau > 0 \mid \text{data})$ across the four ways of measuring university funding. The second column lists the sample size N and number of countries in brackets.

University Funding Measure (+10-Year Horizon)	Waves (Countries)	Total Tax / GDP (pp)	Direct Income Tax / GDP (pp)	Direct Income Tax Share (pp)
1. Per-Student Ratio (% GDP/person)	283 (112)	+0.45 ($P = 0.86$) [−0.35, +1.24]	+0.34 ($P = 0.95$) [−0.06, +0.76]	+1.48 ($P = 0.97$) [−0.11, +3.03]
2. Within-Country Per-Student Increase	176 (86)	+0.53 ($P = 0.82$) [−0.53, +1.67]	+0.81 ($P = 1.00$) [+0.29, +1.33]	+2.33 ($P = 0.99$) [+0.29, +4.36]
3. Real PPP Dollars per Student	283 (112)	+0.94 ($P = 0.99$) [+0.09, +1.88]	+0.39 ($P = 0.97$) [−0.01, +0.78]	+0.80 ($P = 0.81$) [−0.91, +2.47]
4. University Budget % of GDP	403 (123)	+0.17 ($P = 0.70$) [−0.59, +0.83]	+0.11 ($P = 0.73$) [−0.27, +0.47]	+0.94 ($P = 0.91$) [−0.37, +2.45]

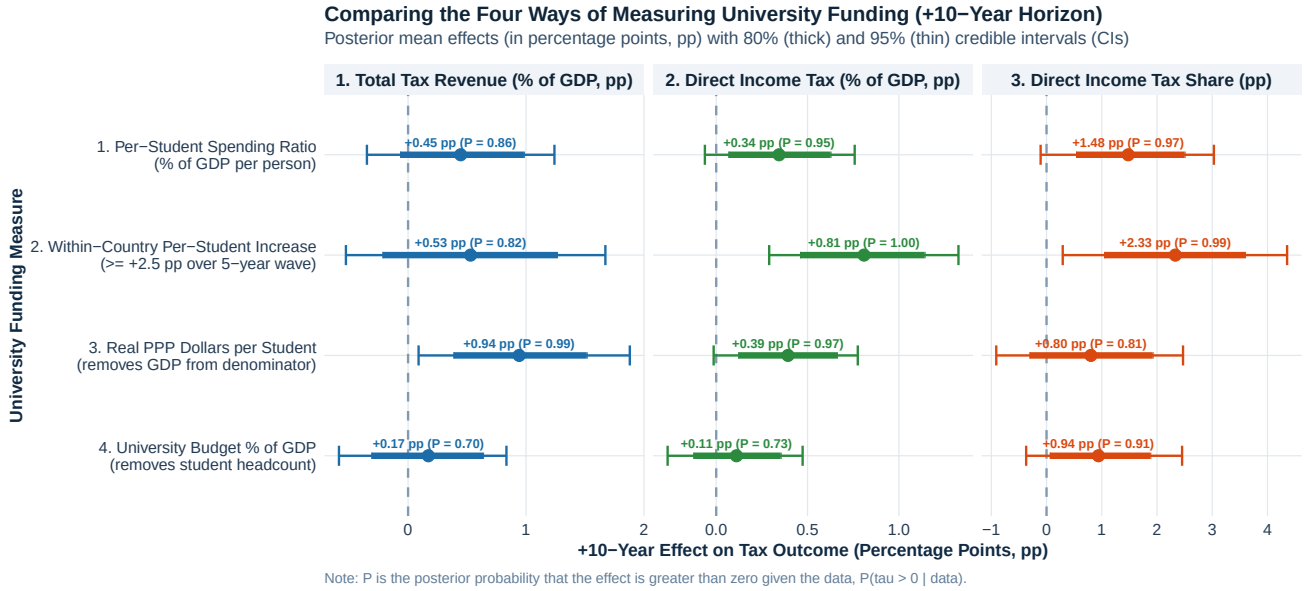


Figure 4: Comparison of +10-year posterior mean effects (in percentage points, pp, with thick 80% and thin 95% CIs) across the four ways of measuring university funding.

The comparison in Table 3 and Figure 4 shows why unbundling the denominator is so informative. When we remove GDP per person from the denominator and measure funding in real 2021 PPP dollars per student (Row 3), the +10-year effect on total non-resource tax revenue more than doubles from +0.45 pp to +0.94 pp of GDP (95% CI $[+0.09, +1.88]$, $P = 0.99$), alongside a +0.39 pp gain in direct income tax as a share of GDP ($P = 0.97$). Even more strikingly, when we focus on country-waves that achieved an actual within-country expansion in per-student funding of at least +2.5 pp over a five-year wave (Row 2, $N = 176$ across 86 countries, representing the top 18.8% of five-year funding increases), the +10-year gain in direct income tax revenue reaches +0.81 pp of GDP (95% CI $[+0.29, +1.33]$, $P = 1.00$) and the direct income tax share rises by +2.33 pp (95% CI $[+0.29, +4.36]$, $P = 0.99$). We also check in Appendix Table 6 (`R13`) what happens when we tighten the static income-group cutoff from the median (top 50%) to the top tercile (top 33%): total non-resource tax revenue at +10 years rises to +0.54 pp of GDP ($P = 0.91$, 80% CI $[+0.02, +1.06]$), whilst direct income tax revenue is +0.24 pp of GDP ($P = 0.88$) and the direct income tax share is +0.87 pp ($P = 0.85$), as placing moderately above-average spenders between the 50th and 67th percentiles into the control group slightly narrows the static cross-sectional contrast compared with the dynamic within-country expansion in Row 2. By contrast, when we remove student headcount from the denominator and look only at the aggregate university budget as a share of

GDP without accounting for how many students share that budget (Row 4), the +10-year effects on total tax revenue (+0.17 pp, $P = 0.70$) and direct income tax revenue (+0.11 pp, $P = 0.73$) are much weaker, consistent with microeconomic evidence that expanding student headcounts without protecting per-student resources leads to institutional crowding and lower completion rates (Bound et al., 2010; Deming and Walters, 2017; Chakrabarti et al., 2020).

5.3 Variation Across Countries, Negative Scenarios, and Country Trajectories

How do the fiscal effects vary across countries at different stages of university enrolment and economic development, and are there situations where increasing university funding actually reduces the tax take? Figure 5 plots the unit-level +10-year posterior effects on the direct income tax share against starting gross tertiary enrolment across all four World Bank income groups, highlighting notable country-waves at the extremes of the data (with full numerical subgroup breakdowns reported in Appendix Table 4). As shown in Figure 5, the shift in the tax mix toward direct income and profit taxation increases steadily with baseline university enrolment and tax administration capacity: it averages +0.90 pp ($P = 0.83$) in elite university systems with under 20% enrolment, rises to +1.53 pp ($P = 0.96$) in massifying systems with 20–50% enrolment (such as Brazil and Armenia), and reaches +1.87 pp (95% CI [+0.16, +3.63], $P = 0.99$) in high-participation systems with at least 50% enrolment, where individual country-waves such as Ireland (2005–2009, +3.01 pp), Austria (2010–2014, +2.72 pp), and the United States (2010–2014, +2.67 pp) achieve the largest compositional shifts. By contrast, when we look at total tax revenue as a share of GDP (Appendix Table 4), the largest gains occur in lower-middle-income (+0.79 pp of GDP, $P = 0.94$) and upper-middle-income (+0.61 pp of GDP, $P = 0.93$) economies whose formal tax bases are still expanding, tapering off to +0.21 pp of GDP ($P = 0.68$) in mature high-income economies.

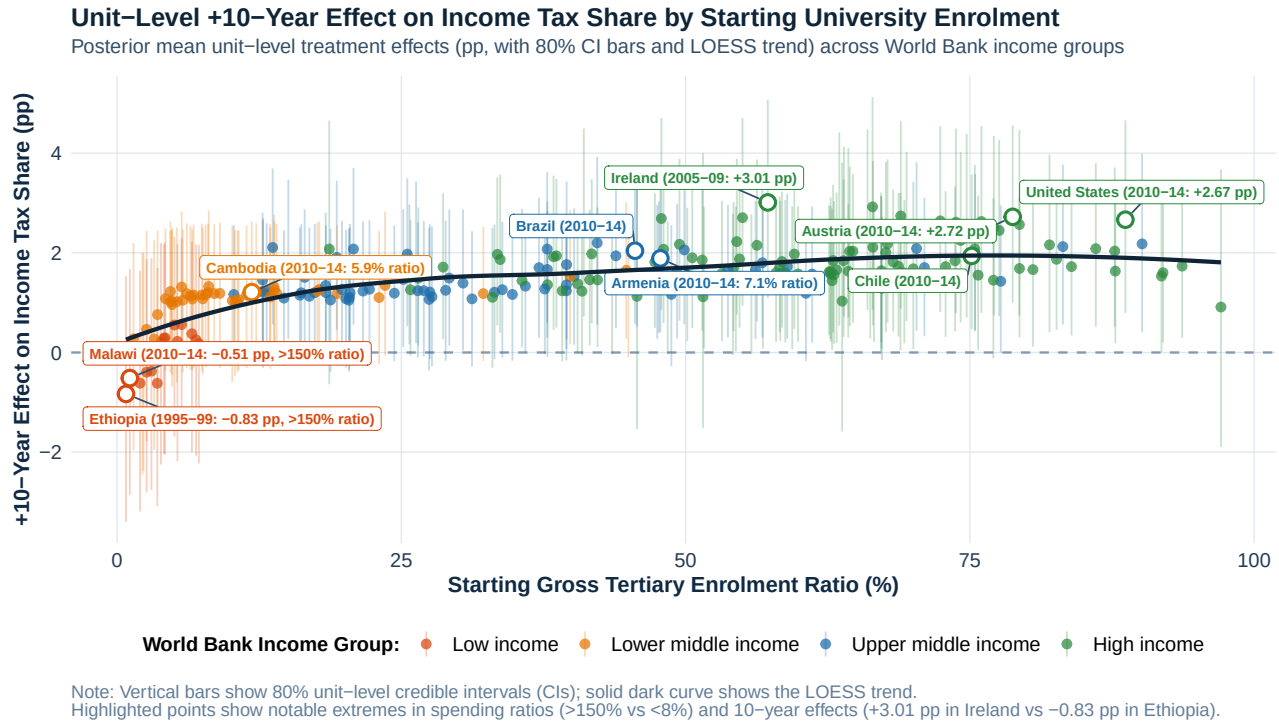


Figure 5: Unit-level +10-year posterior effects on the direct income and profit tax share (in percentage points, pp, with vertical 80% credible interval bars and LOESS trend curve) plotted against starting gross tertiary enrolment across World Bank income groups, highlighting notable countries at the extremes.

Crucially, both the historical data and our model estimates also reveal three specific scenarios where increasing university funding can be associated with a *decrease* in the tax take. First, in the short run (+5 years), 126 out of 372 country-waves (34%) have negative point estimates on total tax-to-GDP (down to −0.57 pp of GDP), reflecting a temporary "student lock-in" effect: when a government expands university funding and student places, more young adults remain in lecture theatres rather than working full time and paying income and consumption taxes during those initial five years. Second, at the +10-year horizon

in very low-income economies with tiny elite enrolment below 4% and extreme UNESCO per-student ratios above 150% (highlighted in the bottom-left corner of Figure 5, such as Ethiopia in `1995-1999` at −0.83 pp and Malawi in `2010-2014` at −0.51 pp), increasing spending on a tiny elite university sector without broad formal-sector job creation or protection against skilled emigration (Beine et al., 2008) fails to expand the domestic income tax base. Third, in mature high-income economies where total tax revenue already exceeds 35% of GDP (and in our separate OECD-only sub-model in Appendix Table 4, where total tax-to-GDP estimates average −0.63 to −0.72 pp), expanding public university budgets can coincide with a lower overall tax-to-GDP ratio even as the income tax share of revenue rises; in Ireland's `2005-2009` wave, for example, high per-student university spending was followed over the next decade by a +3.01 pp shift toward direct income tax share, but headline non-resource tax-to-GDP fell by −6.9 pp (with a model CATE of −0.41 pp) due to the post-2008 banking crisis and multinational national-accounts shifts.

To illustrate how the estimated effect on the direct income tax share evolves from +5 years to +10 years across individual countries, Figure 6 plots the MV-BCF country trajectories and 80% credible intervals for seven illustrative economies alongside the global 80% and 95% posterior credible fan. As shown in Figure 6, countries start with closely grouped effects at the +5-year mark (clustering around +1.26 to +1.32 pp) before diverging by the +10-year mark as university graduates enter different national tax systems. High-participation economies with deep corporate and personal income tax systems show the strongest 10-year maturation on the direct income tax share, led by the United States (+2.67 pp, 80% CI $[+0.79, +4.66]$), Norway (+2.47 pp, $[+0.66, +4.40]$), Austria (+2.27 pp, $[+0.69, +3.99]$), and Ireland (+2.23 pp across all waves, $[+0.60, +3.94]$), whereas Chile (+1.52 pp, $[+0.16, +2.96]$), the United Kingdom (+1.48 pp, $[+0.10, +2.93]$), and South Africa (+1.35 pp, $[-0.03, +2.81]$) exhibit more moderate 10-year shifts after strong initial 5-year gains. In Appendix B, we also summarise our extensive sensitivity checks across alternative estimators, non-overlapping decadal waves, pre-trend controls, two-stage sequential g-estimation, pre-COVID samples, and the top-tercile funding threshold; those checks confirm that while aggregate tax-to-GDP estimates are sensitive to country fixed effects and pandemic-era shocks, the shift toward direct income taxation holds across every specification.

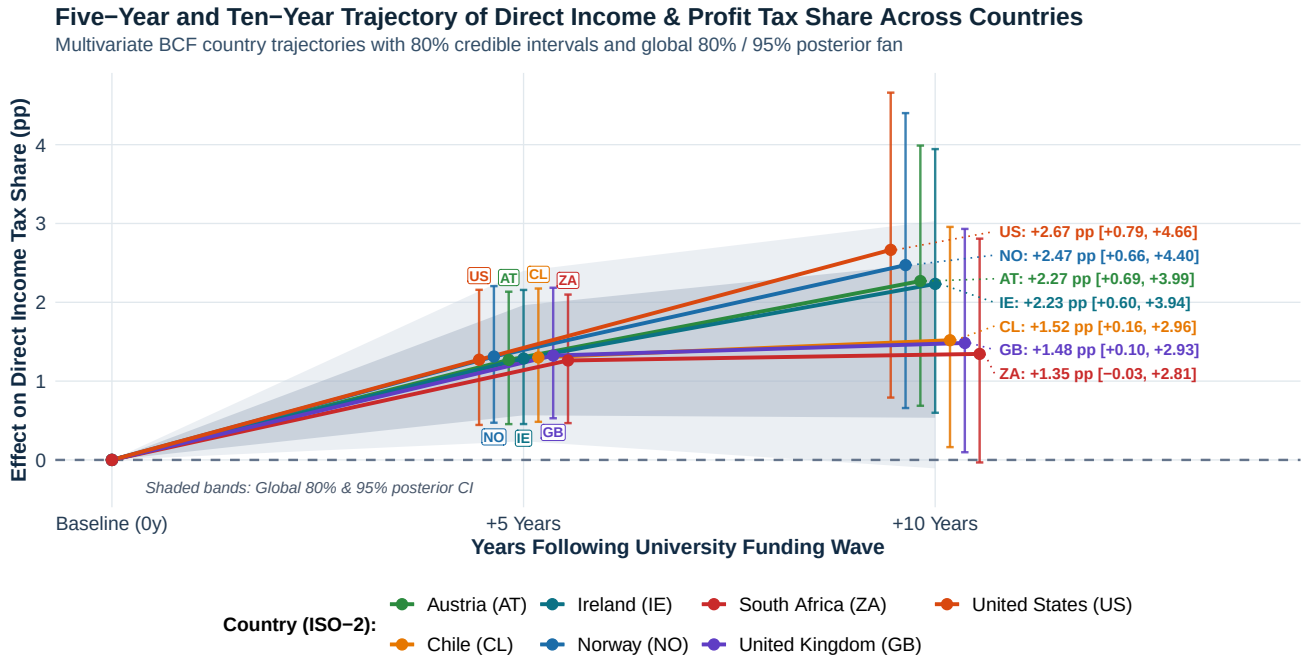


Figure 6: Five-year and 10-year trajectories of the direct income and profit tax share effect across seven illustrative countries from the Multivariate Bayesian Causal Forest, showing country-specific 80% credible intervals (vertical bars, horizontally dodged at +5 and +10 years for visibility) and the global average 80% (dark grey) and 95% (light grey) posterior credible fan.

6 Conclusions and Discussion

6.1 Cohort Maturation Timelines and Policy Implications

Our findings across 121 countries show that government spending on university education produces a measurable fiscal return, but that this return operates primarily by shifting the tax structure toward direct income, profit, and capital gains taxation and requires a full 10-year horizon to mature. Because our panel averages annual data into five-year waves, a university funding increase in wave t (years 0 to 4) is evaluated in wave $t+1$ (years 5 to 9) and wave $t+2$ (years 10 to 14). During the first five-year window, higher per-student university funding immediately expands formal university payrolls (whose academic and administrative staff pay income tax via automatic employer withholding) whilst keeping a larger cohort of young adults in full-time study, which shifts the tax mix toward direct income tax (+1.24 pp) without raising overall tax-to-GDP (+0.02 pp). By the 10-year window (years 10 to 14), two university cohorts have completed their degrees and moved into salaried professional, managerial, and technical roles inside formal firms (Kleven et al., 2016; Jensen, 2022; Valero and Van Reenen, 2019), raising direct income tax revenue by +0.34 to +0.81 pp of GDP and total non-resource tax revenue by +0.45 to +0.94 pp of GDP.

For finance and education ministries, our comparison of the four funding definitions in Section 5.2 carries a clear practical lesson: expanding the overall university budget or student numbers without protecting funding *per student* yields weak fiscal returns (+0.11 pp on direct income tax/GDP when only budget share of GDP is considered). Countries that maintain or expand real resources per student avoid institutional crowding (Bound et al., 2010; Deming and Walters, 2017; Chakrabarti et al., 2020) and achieve much stronger 10-year gains in direct income tax receipts (+0.39 pp of GDP under real PPP dollars per student and +0.81 pp of GDP following observed within-country per-student expansions). To put these magnitudes in perspective using Table 1, median public university budgets stand at 0.47% to 0.94% of GDP across income groups, meaning that a 25% to 35% expansion in per-student funding represents roughly 0.15 to 0.30 pp of GDP in annual public spending. In national currency terms (holding population and baseline GDP constant), for a median-sized economy of €183 billion GDP, an upfront per-student funding boost of 0.15 to 0.30 pp of GDP costs roughly €275 million to €550 million per year and generates an estimated €0.62 billion to €1.48 billion per year in additional direct income tax revenue after 10 years, providing macroeconomic support for microeconomic lifecycle analyses (Trostel, 2010; Hendren and Sprung-Keyser, 2020), though formal net fiscal accounting requires discounting annual cost and revenue streams over complete graduate careers.

6.2 Limitations and Future Research

Several caveats apply when interpreting observational cross-country macro panels, and we highlight five main limitations of our study:

1. *Reverse causality and fiscal-capacity feedback:* A central concern in macro-fiscal studies is that growing tax capacity enables governments to spend more on higher education rather than the other way round (Baumol, 1967; World Bank, 2024; World Bank and UNESCO, 2024). Whilst we adjust for starting tax-to-GDP, starting income and goods tax shares, lagged GDP growth, and explicit pre-treatment fiscal trends (Appendix Table 6), observational conditioning cannot completely rule out multi-decade fiscal feedback loops.

2. *Unobserved concurrent policy reforms:* Our causal estimates rely on conditional ignorability given the 19 baseline characteristics and four follow-up growth controls (Imai et al., 2010; Acharya et al., 2016). If a reformist government simultaneously modernises its electronic tax administration (Pomeranz and Vila-Belda, 2019; Okunogbe and Pouliquen, 2022) and increases university funding within the same five-year wave, part of the estimated effect could reflect that concurrent tax reform.

3. *Skilled emigration (brain drain):* In low- and lower-middle-income countries, a substantial share of university graduates emigrate to higher-income economies (Beine et al., 2008). Because emigrants

pay direct income taxes in their destination country whilst sending untaxed remittances home (which are spent on domestic goods subject to VAT), high skilled emigration can dampen domestic direct income tax returns in smaller developing nations.

4. *Central government reporting and five-year wave coarseness:* World Bank fiscal series primarily record central government revenue, which can understate state or provincial income taxes in federal countries, and averaging annual data into five-year waves smooths out year-to-year timing.

5. *Binary treatment threshold and boundary effects:* Because our primary treatment indicator dichotomises per-student funding at each income group's median ($Z_{i,t} \in \{0, 1\}$), it estimates the average effect of shifting from a typical below-median funding level (a median ratio of 22.2% in high-income $Z = 0$ countries) to a typical above-median funding level (36.2% in high-income $Z = 1$ countries). In the most recent data wave (`2015-2019`), several $Z = 0$ countries sit almost exactly on their income-group boundary, such as Portugal (26.91% vs the 26.95% high-income cutoff, a gap of just 0.04 pp costing €0.004 billion per year) and Poland (26.77%, a gap of 0.18 pp costing €0.12 billion per year). Interpreting a binary threshold model literally as a step function would imply that a token spending nudge across the median line triggers the full 10-year tax return (€0.98 billion per year in Portugal or €2.84 billion per year in Poland), which would clearly be unrealistic. As we detail in Appendix Table 7, policy makers in near-boundary countries should evaluate the cost of a substantive per-student funding expansion (such as the +2.5 pp within-country expansion in Row 2 of Table 3) rather than the minimal cost of stepping over the median cutoff.

These limitations point to several promising avenues for future research. First, whilst this paper has focussed exclusively on tertiary education because university cohorts enter the labour market within a 5-to-10-year window, extending multivariate Bayesian causal forests to 15-to-25-year horizons would allow researchers to compare government spending on early childhood, primary, and secondary schooling (Hanushek and Woessmann, 2012; Angrist et al., 2021; Hendren and Sprung-Keyser, 2020) directly against university funding as younger school cohorts reach adulthood. Second, incorporating bilateral skilled migration data (Beine et al., 2008) directly into the treatment-effect moderators would quantify how brain drain alters the fiscal return to public university subsidies in developing countries. Third, linking university funding cohorts to administrative employer-employee tax registers within individual countries would make it possible to trace graduate earnings and corporate tax payments year by year. To help policy makers and researchers explore our dataset and country-level estimates interactively, we provide a companion web application in the `app/` directory of the project repository.

Declarations and Statements

Author Contributions (CRediT). **Andrew C. Parnell:** Conceptualisation, Data curation, Formal analysis, Methodology, Software, Visualisation, Writing – original draft, Writing – review and editing.

Data Availability Statement. All raw and processed datasets analysed in this study are derived from publicly available international databases (World Bank WDI, UNESCO UIS, IMF, WGI, SWIID, and OECD) licensed under CC-BY-4.0 and are included in the project repository at `https://github.com/andrewcparnell/education-tax-bcf` (with full SHA-256 checksums and provenance documented in `data/raw/PROVENANCE.md`).

Code Availability Statement. All R and C++ (`RcppArmadillo`) source code, data-cleaning scripts, MCMC samplers, diagnostic scripts, interactive exploration app (`app/`), automated pipeline tests, and figure generation pipelines are openly available under the MIT License at `https://github.com/andrewcparnell/education-tax-bcf` and can be reproduced end-to-end via `bash run_all.sh`.

Ethics, Funding, and Conflict of Interest. This study analyses aggregate sovereign macroeconomic statistics and involves no human subjects or personally identifiable information; institutional ethical

approval is not required. This research received no specific external grant funding. As disclosed in Section 2, the author co-developed the Multivariate Bayesian Causal Forest methodology (McJames et al., 2025) evaluated alongside competing estimators in this study; the author declares no financial or other competing interests.

A Mathematical Specification of the MV-BCF Sampler

Let $Y_{i,t+h} = (Y^{(1)}_{i,t+h}, Y^{(2)}_{i,t+h}, Y^{(3)}_{i,t+h}, Y^{(4)}_{i,t+h})^\top \in \mathbb{R}^4$ denote the vector of four disaggregated percentage-point fiscal changes for country i over horizon $h \in \{1, 2\}$ (+5 and +10 years). Following Hahn et al. (2020) and McJames et al. (2025), the Multivariate Bayesian Causal Forest (MV-BCF) models $Y_{i,t+h}$ as:

$$Y_{i,t+h} = \sum_{m=1}^{M_\mu} g_\mu(X_{i,t}, \hat{\pi}_{i,t}; T^\mu_m, M^\mu_m) + Z_{i,t} \sum_{l=1}^{M_\tau} g_\tau(W_{i,t}; T^\tau_l, M^\tau_l) + \varepsilon_{i,t+h}, \qquad \varepsilon_{i,t+h} \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(0, \Sigma), \quad (3)$$

where the prognostic forest uses $M_\mu = 75$ trees with depth-penalty hyperparameters ($\alpha_\mu = 0.95, \beta_\mu = 2.0$) and the treatment-effect forest uses $M_\tau = 30$ trees with more conservative depth-penalty hyperparameters ($\alpha_\tau = 0.25, \beta_\tau = 3.0$) (Hahn et al., 2020). Each prognostic leaf vector $\mu_{m,b} \in \mathbb{R}^4$ and treatment leaf vector $\tau_{l,b} \in \mathbb{R}^4$ receives an independent multivariate Gaussian prior $\mathcal{N}_4(0, \Sigma_\mu)$ and $\mathcal{N}_4(0, \Sigma_\tau)$ scaled inversely by the number of trees. Given partial residuals $R^{(m)}_i$ and precision matrix Σ^{-1}, the conjugate leaf posterior is Gaussian with covariance $\Omega_b = (\Sigma_\mu^{-1} + n_b \Sigma^{-1})^{-1}$ and mean $\Omega_b \Sigma^{-1} \sum_{i \in b} R^{(m)}_i$. During reversible-jump Metropolis-Hastings tree updates (Chipman et al., 2010), a `GROW` move on a terminal node η at depth d_η (with splitting probability $p_{\text{split}}(\eta) = \alpha(1 + d_\eta)^{-\beta}$) is accepted with probability $\min(1, r_{\text{grow}})$, where r_{grow} combines the closed-form Gaussian marginal likelihood ratio with the exact tree prior and transition ratio:

$$r_{\text{grow}} = \frac{p_{\text{prune}}\, b(T)}{p_{\text{grow}}\, w(T^*)} \cdot \frac{p_{\text{split}}(\eta)\big(1 - p_{\text{split}}(\eta_L)\big)\big(1 - p_{\text{split}}(\eta_R)\big)}{1 - p_{\text{split}}(\eta)} \cdot \frac{L(R \mid T^*, \Sigma)}{L(R \mid T, \Sigma)}, \quad (4)$$

where $b(T)$ is the number of terminal nodes in T and $w(T^*)$ is the number of singly internal (`nog`) parent nodes in the proposed tree T^* (with the exact inverse ratio used for `PRUNE` moves). Given full residuals $E_i = Y_{i,t+h} - \mu_i - \tau_i Z_{i,t}$, the cross-outcome covariance matrix Σ receives a conjugate Inverse-Wishart prior $\mathcal{IW}(\nu_0, \Psi_0)$ with $\nu_0 = 6$ and is updated via $\Sigma \mid E \sim \mathcal{IW}(\nu_0 + N, \Psi_0 + \sum_{i=1}^N E_i E_i^\top)$.

B Subgroup Breakdowns, Method Benchmarks, and Robustness Checks

In this appendix we present and interpret the detailed subgroup estimates, competing estimator benchmarks, and sensitivity checks referenced in Section 5. Table 4 reports the +10-year subgroup posterior means and 95% credible intervals across baseline university enrolment tiers ($N = 244$ observed out of 283), all four World Bank income groups ($N = 283$), and the five OECD financing regimes ($N = 119$ OECD country-waves). Two contrasting lessons emerge from Table 4. First, the compositional shift toward direct income and profit taxation (Column 5) strengthens steadily with university participation and administrative capacity, rising from +0.90 pp ($P = 0.83$) in elite systems ($< 20\%$ enrolment) to +1.53 pp ($P = 0.96$) in massifying systems and +1.87 pp ($P = 0.99$) in high-participation systems ($\geq 50\%$ enrolment), and remaining consistently positive (+1.65 to +1.72 pp, $P = 0.91$–0.92) across all five OECD financing regimes regardless of tuition structure. Second, total non-resource tax revenue as a share of GDP (Column 3) follows the opposite pattern across development stages: the largest gains occur in lower-middle-income (+0.79 pp, $P = 0.94$) and upper-middle-income (+0.61 pp, $P = 0.93$) economies whose formal tax bases are still expanding, whereas mature high-income economies (+0.21 pp, $P = 0.68$) and the separate OECD-only sub-model (-0.63 to -0.72 pp) already operate near their aggregate tax-to-GDP ceiling and primarily experience a shift in the tax mix toward direct income taxation.

Table 4: Subgroup +10-year posterior mean effects (in percentage points, pp, with 95% CIs on the second line of each cell and posterior probabilities $P = P(\tau > 0 \mid \text{data})$) across starting university enrolment tiers ($N = 244$ observed), World Bank income groups ($N = 283$), and OECD financing regimes ($N = 119$ OECD waves). Subgroups with $N \leq 20$ are small-sample exploratory strata.

Subgroup (+10-Year Horizon)	Waves (Countries)	Total Tax / GDP (pp)	Direct Income Tax / GDP (pp)	Direct Income Tax Share (pp)
1. Elite Enrolment ($< 20\%$)	74 (43)	$+0.77$ ($P = 0.93$) $[-0.22, +1.79]$	$+0.47$ ($P = 0.97$) $[+0.003, +0.96]$	$+0.90$ ($P = 0.83$) $[-1.07, +2.76]$
2. Massifying (20–50%)	73 (42)	$+0.52$ ($P = 0.90$) $[-0.32, +1.33]$	$+0.35$ ($P = 0.94$) $[-0.07, +0.76]$	$+1.53$ ($P = 0.96$) $[-0.16, +3.18]$
3. High-Participation ($\geq 50\%$)	97 (44)	$+0.22$ ($P = 0.68$) $[-0.77, +1.19]$	$+0.25$ ($P = 0.85$) $[-0.23, +0.71]$	$+1.87$ ($P = 0.99$) $[+0.16, +3.63]$
4. Low Income (exploratory)	15 (9)	$+0.86$ ($P = 0.88$) $[-0.54, +2.18]$	$+0.56$ ($P = 0.97$) $[-0.01, +1.23]$	-0.03 ($P = 0.54$) $[-3.31, +2.67]$
5. Lower Middle Income	47 (21)	$+0.79$ ($P = 0.94$) $[-0.23, +1.86]$	$+0.46$ ($P = 0.97$) $[-0.02, +0.96]$	$+1.06$ ($P = 0.86$) $[-0.88, +3.04]$
6. Upper Middle Income	76 (35)	$+0.61$ ($P = 0.93$) $[-0.22, +1.46]$	$+0.35$ ($P = 0.94$) $[-0.09, +0.78]$	$+1.50$ ($P = 0.95$) $[-0.26, +3.23]$
7. High Income	145 (47)	$+0.21$ ($P = 0.68$) $[-0.75, +1.09]$	$+0.28$ ($P = 0.88$) $[-0.18, +0.74]$	$+1.76$ ($P = 0.99$) $[+0.12, +3.55]$
8. OECD Nordic High-Public	20 (5)	-0.72 ($P = 0.03$) $[-1.46, +0.04]$	-0.01 ($P = 0.50$) $[-0.68, +0.63]$	$+1.72$ ($P = 0.92$) $[-0.69, +4.11]$
9. OECD Low-Tuition Public	46 (15)	-0.70 ($P = 0.03$) $[-1.41, +0.03]$	-0.01 ($P = 0.51$) $[-0.69, +0.62]$	$+1.69$ ($P = 0.92$) $[-0.49, +4.06]$
10. OECD Hybrid Public-Fee	29 (8)	-0.69 ($P = 0.03$) $[-1.38, +0.05]$	-0.03 ($P = 0.48$) $[-0.71, +0.60]$	$+1.67$ ($P = 0.92$) $[-0.58, +3.99]$
11. OECD High-Tuition High-Sub.	11 (5)	-0.69 ($P = 0.04$) $[-1.45, +0.08]$	-0.03 ($P = 0.48$) $[-0.70, +0.62]$	$+1.67$ ($P = 0.92$) $[-0.71, +4.06]$
12. OECD High-Tuition Low-Sub.	13 (4)	-0.63 ($P = 0.04$) $[-1.45, +0.10]$	$+0.01$ ($P = 0.53$) $[-0.65, +0.66]$	$+1.65$ ($P = 0.91$) $[-0.76, +4.05]$

Table 5 compares our primary joint four-outcome MV-BCF model at the +10-year horizon ($N = 283$) against a specification that omits the follow-up macroeconomic growth controls (Total Effect), three Bayesian tree ablations (B1 univariate BCFs, B2 MV-BCF without the propensity score $\hat{\pi}(X)$, and B3 single-forest BART), and three frequentist panel estimators (B4 linear local projections, Jordà 2005; B5 country-and-wave two-way fixed effects; and B6 5-fold country-blocked cross-fitted double machine learning, Chernozhukov et al. 2018). Three methodological lessons stand out from Table 5. First, comparing the first two rows shows that omitting the follow-up macroeconomic growth controls leaves the estimated effects virtually unchanged (+0.45 vs +0.45 pp on total tax/GDP and +1.48 vs +1.36 pp on the direct income tax share), confirming that our findings are not driven by conditioning on post-treatment economic growth. Second, amongst the Bayesian tree specifications (B1–B3), univariate BCFs (B1) and MV-BCF without the propensity score (B2) yield estimates closely aligned with our primary model, whereas single-forest BART (B3) shrinks the treatment effects toward zero (+0.17 pp on direct income tax/GDP and +1.05 pp on the income tax share), illustrating the regularisation-induced confounding described by Hahn et al. (2020) when prognostic and treatment trees are not given separate priors. Third, comparing across the frequentist panel estimators (B4–B6) shows that whilst headline total tax-to-GDP is sensitive to country fixed effects (B5, −0.26 pp vs +0.80 pp in linear local projections B4 and +1.13 pp in double machine learning B6), both direct income tax/GDP (+0.29 to +0.40 pp) and the direct income tax share (+1.19 to +1.55 pp) remain consistently positive and of similar magnitude across every Bayesian and frequentist method.

Table 5: Method benchmark and ablation comparison at the +10-year horizon on the identical common sample ($N = 283$, 112 countries). Posterior means, 95% Bayesian CIs, and posterior probabilities P are shown for the Bayesian tree specifications; frequentist point estimates, 95% confidence intervals, and one-sided p-value complements $P_{\text{freq}} = 1 - p_{\text{one-sided}}$ are shown for B4–B6.

Estimator / Specification (+10y)	Total Tax / GDP (pp)	Direct Income Tax / GDP (pp)	Direct Income Tax Share (pp)	Goods Tax Share (pp)
Primary: Joint MV-BCF (Full Controls)	+0.45 ($P = 0.86$) [−0.35, +1.24]	+0.34 ($P = 0.95$) [−0.06, +0.76]	+1.48 ($P = 0.97$) [−0.11, +3.03]	−0.56 ($P = 0.24$) [−2.02, +1.05]
MV-BCF Without Growth Controls (Total)	+0.45 ($P = 0.87$) [−0.29, +1.21]	+0.25 ($P = 0.88$) [−0.16, +0.67]	+1.36 ($P = 0.96$) [−0.21, +2.92]	−0.55 ($P = 0.25$) [−2.11, +0.96]
B1: Univariate BCFs ($D = 1$)	+0.57 ($P = 0.91$) [−0.31, +1.37]	+0.39 ($P = 0.97$) [−0.01, +0.78]	+1.60 ($P = 0.97$) [−0.02, +3.28]	−0.32 ($P = 0.37$) [−2.14, +1.49]
B2: MV-BCF Without $\hat{\pi}(X)$	+0.45 ($P = 0.87$) [−0.31, +1.23]	+0.28 ($P = 0.93$) [−0.09, +0.64]	+1.54 ($P = 0.98$) [+0.09, +3.04]	−0.54 ($P = 0.25$) [−1.96, +0.97]
B3: Single-Forest BART	+0.28 ($P = 0.80$) [−0.34, +0.91]	+0.17 ($P = 0.81$) [−0.15, +0.53]	+1.05 ($P = 0.94$) [−0.14, +2.46]	−0.38 ($P = 0.30$) [−1.92, +1.02]
B4: Linear Local Projections	+0.80 ($P_{\text{freq}} = 0.95$) [−0.18, +1.79]	+0.29 ($P_{\text{freq}} = 0.87$) [−0.21, +0.79]	+1.19 ($P_{\text{freq}} = 0.91$) [−0.56, +2.94]	+0.72 ($P_{\text{freq}} = 0.80$) [−0.97, +2.41]
B5: Two-Way Fixed Effects	−0.26 ($P_{\text{freq}} = 0.33$) [−1.40, +0.88]	+0.34 ($P_{\text{freq}} = 0.84$) [−0.34, +1.01]	+1.55 ($P_{\text{freq}} = 0.92$) [−0.60, +3.69]	−0.13 ($P_{\text{freq}} = 0.46$) [−2.79, +2.54]
B6: Cross-Fitted Double ML	+1.13 ($P_{\text{freq}} = 0.99$) [+0.21, +2.05]	+0.40 ($P_{\text{freq}} = 0.96$) [−0.06, +0.85]	+1.34 ($P_{\text{freq}} = 0.95$) [−0.30, +2.98]	+0.50 ($P_{\text{freq}} = 0.70$) [−1.34, +2.34]

Table 6 reports 14 sensitivity and falsification checks (R1 to R13) designed to test every data-construction and modelling decision in our pipeline. First, the +10-year gains in direct income tax revenue and direct income tax share remain strongly positive when we use raw unadjusted WDI tax-to-GDP without natural resource rent deductions (R1, +0.30 and +1.48 pp), collapse the panel into non-overlapping 10-year decadal waves (R2, +0.36 and +1.96 pp, $P = 0.98$), drop all observations with any missing baseline covariate (R4, +0.38 and +1.94 pp, $P = 0.99$), use a more diffuse treatment-effect tree prior (R6), perturb SWIID Gini inequality with its reported standard errors (R7), use 5-fold country-blocked out-of-fold propensity scores (R8), apply two-stage sequential g-estimation to demediate post-treatment growth (R9; Acharya et al. 2016, +0.39 and +1.51 pp, $P = 0.99$), control explicitly for lagged pre-treatment tax trends $\Delta Y_{t-1\to t}$ (R10, +0.32 and +1.68 pp, $P = 0.98$), or replace the goods tax share with goods tax as a percentage of GDP (R11). Second, our pre-treatment falsification test (R5), which checks whether university funding at wave t "predicts" prior tax changes over $t-1 \to t$, yields small and statistically indistinct effects (+0.07 pp on direct income tax/GDP, $P = 0.66$, with 95% CIs spanning zero across

all outcomes), reassuring us that treatment assignment is not simply tracking pre-existing fiscal booms. Third, the three most restrictive subsample and threshold checks (within-country demeaning at +10 years in `R3b`, restricting the sample to pre-COVID waves before 2020 in `R12`, which drops the `2020-2024` outcome wave and reduces the sample to $N = 187$, and tightening the static threshold to the top tercile of each income group in `R13`) preserve positive point estimates across all three tax measures (with total tax/GDP reaching +0.71 pp, $P = 0.92$ in `R12` and +0.54 pp, $P = 0.91$ in `R13`), although the smaller effective number of within-country treatment transitions widens the credible intervals on the direct income tax outcomes.

Table 6: Robustness, identification, and falsification checks across the primary tax outcomes (with 95% CIs on the second line of each cell and posterior probabilities $P = P(\tau > 0 \mid \text{data})$).

Sensitivity Check	Horizon (N, Countries)	Total Tax / GDP (pp)	Direct Income Tax / GDP (pp)	Direct Income Tax Share (pp)
`R1.` **Unadjusted WDI Tax / GDP**	+10y (283, 112)	+0.34 ($P = 0.79$) [−0.42, +1.08]	+0.30 ($P = 0.94$) [−0.10, +0.70]	+1.48 ($P = 0.97$) [−0.04, +3.01]
`R2.` **Non-Overlapping 10y Waves**	+10y (154, 105)	+0.64 ($P = 0.91$) [−0.31, +1.56]	+0.36 ($P = 0.93$) [−0.10, +0.81]	+1.96 ($P = 0.98$) [+0.09, +3.92]
`R3a.` **Country Fixed Effects (+5y)**	+5y (372, 121)	+0.08 ($P = 0.62$) [−0.43, +0.56]	+0.19 ($P = 0.93$) [−0.06, +0.43]	+0.96 ($P = 0.97$) [−0.06, +2.00]
`R3b.` **Country Fixed Effects (+10y)**	+10y (283, 112)	+0.32 ($P = 0.82$) [−0.34, +0.97]	+0.15 ($P = 0.79$) [−0.20, +0.51]	+0.69 ($P = 0.84$) [−0.63, +2.06]
`R4.` **Strictly Observed Covariates**	+10y (238, 107)	+0.56 ($P = 0.92$) [−0.22, +1.30]	+0.38 ($P = 0.98$) [+0.01, +0.81]	+1.94 ($P = 0.99$) [+0.36, +3.60]
`R5.` **Placebo ($\Delta Y_{t-1\to t}$)**	Pre-t (348, 117)	+0.24 ($P = 0.79$) [−0.34, +0.79]	+0.07 ($P = 0.66$) [−0.21, +0.35]	+0.59 ($P = 0.83$) [−0.55, +1.68]
`R6.` **Diffuse τ Prior**	+10y (283, 112)	+0.47 ($P = 0.89$) [−0.27, +1.19]	+0.29 ($P = 0.91$) [−0.13, +0.71]	+1.67 ($P = 0.98$) [+0.03, +3.16]
`R7.` **SWIID Error Perturbation**	+10y (283, 112)	+0.55 ($P = 0.90$) [−0.27, +1.38]	+0.26 ($P = 0.90$) [−0.14, +0.66]	+1.49 ($P = 0.97$) [−0.01, +2.99]
`R8.` **Country-Blocked OOF PS**	+10y (283, 112)	+0.55 ($P = 0.93$) [−0.15, +1.28]	+0.26 ($P = 0.91$) [−0.14, +0.64]	+1.49 ($P = 0.98$) [+0.04, +2.94]
`R9.` **Sequential G-Estimation**	+10y (283, 112)	+0.70 ($P = 0.97$) [−0.02, +1.47]	+0.39 ($P = 0.97$) [−0.002, +0.78]	+1.51 ($P = 0.99$) [+0.18, +2.97]
`R10.` **Pre-Trend Controls ($\Delta Y_{t-1\to t}$)**	+10y (283, 112)	+0.44 ($P = 0.85$) [−0.37, +1.21]	+0.32 ($P = 0.93$) [−0.08, +0.72]	+1.68 ($P = 0.98$) [+0.11, +3.14]
`R11.` **Goods Tax / GDP Check**	+10y (283, 112)	+0.54 ($P = 0.92$) [−0.23, +1.33]	+0.30 ($P = 0.94$) [−0.09, +0.73]	+1.42 ($P = 0.96$) [−0.21, +3.13]
`R12.` **Pre-COVID Sample (< 2020)**	+10y (187, 90)	+0.71 ($P = 0.92$) [−0.30, +1.75]	+0.17 ($P = 0.71$) [−0.35, +0.70]	+0.74 ($P = 0.77$) [−1.30, +2.71]
`R13.` **Top-Tercile Cutoff (Top 33%)**	+10y (283, 112)	+0.54 ($P = 0.91$) [−0.26, +1.35]	+0.24 ($P = 0.88$) [−0.16, +0.63]	+0.87 ($P = 0.85$) [−0.64, +2.36]

Finally, Table 7 lists all 48 countries from our 10-year modelling sample that sit in the below-median group ($Z_{i,t} = 0$) in the most recent treatment wave (`2015-2019`), ordered within each World Bank income group from closest to the cutoff to furthest below it (full data in `outputs/tables/tableA4_latest_z0_countries_costs_and_returns.csv`). Table 7 highlights two practical insights for policy makers. First, eight countries (marked with *: Portugal, Poland, New Zealand, Iceland, Uruguay, Romania, Italy, and Malaysia) sit within 2.5 pp of their income-group median cutoff (26.9% for high income and 26.8% for upper-middle income); for these near-boundary countries, the literal cost of stepping across the median line is very small (€0.004 billion per year in Portugal and €0.12 billion per year in Poland), so Column 3 also reports the annual cost of a substantive +2.5 pp per-student funding expansion (€0.32 billion in Portugal and €1.49 billion in Poland, corresponding to the within-country expansion benchmark in Row 2 of Table 3). Second, for countries sitting further below their income-group median (such as Ireland at 17.3% needing €2.30 billion per year to reach 26.9%, Spain at 22.4% needing €3.83 billion, Chile at 19.6% needing €2.68 billion, Colombia at 20.6% needing €2.56 billion, and Bangladesh at 29.9% needing €10.75 billion to reach 76.5%), moving to $Z_{i,t} = 1$ represents a major fiscal commitment whose annual cost is of the same order of magnitude as the estimated 10-year gain in annual direct income tax receipts (Column

5), whilst every country exhibits a positive expected shift in the direct income tax share (Column 6) alongside wide 95% credible intervals that reflect honest country-level macroeconomic uncertainty.

Table 7: Countries in the below-median funding group ($Z_{i,t} = 0$) in the most recent treatment wave (2015–2019, $N = 48$ countries with +10-year MV-BCF estimates), showing their current per-student spending ratio (% of GDP per person) and gap to the income-group median threshold (26.9% High, 26.8% Upper Mid., 76.5% Lower Mid., 205% Low), annual public cost to reach $Z = 1$ in % of GDP and constant 2021 euros (for near-boundary countries marked * with a gap under 2.5 pp, the second figure shows the cost of a substantive +2.5 pp per-student increase), and estimated +10-year posterior mean returns (with 95% CIs) on direct income tax/GDP (pp), annual direct income tax revenue (€ billions/yr, holding latest population and PPP GDP constant), and the direct income tax share (pp).

Country (Income Group)	Current Ratio (Gap to $Z = 1$)	Annual Cost to Reach $Z = 1$ % GDP (€B/yr / +2.5pp*)	+10y Inc. Tax / GDP (pp) [95% CI]	+10y Annual Inc. Tax Return (€B/yr) [95% CI]	+10y Inc. Tax Share (pp) [95% CI]
Portugal* (High)	26.9% (+0.04 pp)	0.001% (€0.004B / €0.32B*)	+0.25 [−0.31, +0.85]	€+0.98 [−1.19, +3.27]	+1.56 [−0.55, +3.77]
Poland* (High)	26.8% (+0.18 pp)	0.008% (€0.12B / €1.49B*)	+0.20 [−0.45, +0.83]	€+2.84 [−6.51, +11.94]	+1.95 [−0.33, +4.62]
New Zealand* (High)	26.1% (+0.8 pp)	0.04% (€0.10B / €0.29B*)	+0.26 [−0.42, +0.86]	€+0.60 [−0.97, +1.97]	+1.66 [−0.68, +4.40]
Iceland* (High)	26.1% (+0.9 pp)	0.05% (€0.01B / €0.03B*)	+0.20 [−0.50, +0.85]	€+0.05 [−0.11, +0.19]	+2.00 [−0.44, +4.72]
Uruguay* (High)	25.3% (+1.7 pp)	0.07% (€0.07B / €0.10B*)	+0.41 [−0.13, +0.97]	€+0.39 [−0.12, +0.91]	+1.24 [−0.86, +3.38]
Romania* (High)	25.1% (+1.9 pp)	0.05% (€0.37B / €0.49B*)	+0.25 [−0.35, +0.83]	€+1.73 [−2.34, +5.59]	+1.62 [−0.54, +3.87]
Italy* (High)	24.8% (+2.1 pp)	0.07% (€1.82B / €2.12B*)	+0.26 [−0.34, +0.87]	€+7.26 [−9.44, +24.02]	+1.66 [−0.54, +4.27]
Slovenia (High)	24.1% (+2.9 pp)	0.12% (€0.10B)	+0.16 [−0.50, +0.77]	€+0.15 [−0.45, +0.69]	+2.04 [−0.29, +4.87]
Singapore (High)	23.5% (+3.4 pp)	0.16% (€1.10B)	+0.80 [−0.22, +1.91]	€+5.49 [−1.53, +13.15]	+1.48 [−2.90, +5.67]
Hungary (High)	23.3% (+3.6 pp)	0.11% (€0.37B)	+0.22 [−0.38, +0.81]	€+0.76 [−1.31, +2.83]	+1.80 [−0.31, +4.17]
Latvia (High)	22.6% (+4.4 pp)	0.18% (€0.11B)	+0.22 [−0.41, +0.81]	€+0.14 [−0.26, +0.52]	+1.59 [−0.59, +3.92]
Spain (High)	22.4% (+4.6 pp)	0.19% (€3.83B)	+0.18 [−0.44, +0.78]	€+3.66 [−8.83, +15.73]	+1.98 [−0.26, +4.76]
Macao SAR, China (High)	22.3% (+4.7 pp)	0.24% (€0.12B)	+0.73 [−0.22, +1.60]	€+0.37 [−0.11, +0.82]	+1.65 [−1.69, +5.24]
Lithuania (High)	21.4% (+5.5 pp)	0.25% (€0.30B)	+0.23 [−0.39, +0.80]	€+0.27 [−0.46, +0.96]	+1.55 [−0.59, +3.76]
Czechia (High)	20.7% (+6.3 pp)	0.19% (€0.87B)	+0.21 [−0.40, +0.82]	€+0.99 [−1.89, +3.85]	+2.02 [−0.22, +4.79]
United States (High)	20.5% (+6.5 pp)	0.45% (€101.19B)	+0.01 [−0.88, +0.69]	€+1.40 [−196.23, +153.47]	+2.67 [−0.02, +6.12]
Australia (High)	19.6% (+7.3 pp)	0.59% (€8.42B)	+0.22 [−0.37, +0.82]	€+3.10 [−5.29, +11.75]	+1.70 [−0.51, +4.28]
Chile (High)	19.6% (+7.4 pp)	0.52% (€2.68B)	+0.28 [−0.29, +0.85]	€+1.47 [−1.52, +4.45]	+1.52 [−0.57, +3.75]
Russian Federation (High)	19.3% (+7.7 pp)	0.35% (€18.13B)	+0.25 [−0.45, +0.92]	€+12.75 [−23.40, +47.46]	+2.03 [−0.50, +4.73]
Israel (High)	18.7% (+8.2 pp)	0.41% (€1.69B)	+0.36 [−0.19, +0.92]	€+1.49 [−0.79, +3.79]	+1.38 [−0.77, +3.71]
Ireland (High)	17.3% (+9.7 pp)	0.41% (€2.30B)	+0.22 [−0.45, +0.87]	€+1.23 [−2.47, +4.84]	+2.23 [−0.20, +4.98]
Korea, Rep. (High)	14.9% (+12.1 pp)	0.70% (€17.51B)	+0.24 [−0.44, +0.92]	€+6.06 [−10.93, +23.00]	+1.96 [−0.54, +4.65]
Greece (High)	11.2% (+15.8 pp)	1.24% (€4.17B)	+0.21 [−0.43, +0.80]	€+0.69 [−1.44, +2.70]	+1.52 [−0.65, +3.79]
Malaysia* (Upper Mid.)	24.8% (+2.0 pp)	0.08% (€0.83B / €1.02B*)	+0.34 [−0.24, +0.91]	€+3.47 [−2.44, +9.24]	+1.45 [−0.66, +3.78]
Jordan (Upper Mid.)	23.7% (+3.1 pp)	0.09% (€0.09B)	+0.48 [−0.06, +1.03]	€+0.50 [−0.06, +1.08]	+1.11 [−1.18, +3.24]
Indonesia (Upper Mid.)	20.8% (+6.1 pp)	0.07% (€2.47B)	+0.45 [−0.13, +1.03]	€+15.52 [−4.48, +35.46]	+1.21 [−1.15, +3.60]
Colombia (Upper Mid.)	20.6% (+6.2 pp)	0.30% (€2.56B)	+0.35 [−0.19, +0.92]	€+2.95 [−1.63, +7.73]	+1.35 [−0.76, +3.47]
Azerbaijan (Upper Mid.)	20.2% (+6.6 pp)	0.14% (€0.28B)	+0.25 [−0.39, +0.87]	€+0.49 [−0.75, +1.68]	+1.99 [−0.31, +4.57]
Guatemala (Upper Mid.)	18.1% (+8.8 pp)	0.21% (€0.41B)	+0.43 [−0.10, +0.96]	€+0.85 [−0.20, +1.91]	+1.20 [−0.92, +3.30]
Belarus (Upper Mid.)	17.0% (+9.9 pp)	0.47% (€1.09B)	+0.25 [−0.43, +0.87]	€+0.60 [−1.00, +2.05]	+2.00 [−0.51, +4.70]
Argentina (Upper Mid.)	16.7% (+10.1 pp)	0.67% (€7.37B)	+0.40 [−0.19, +0.96]	€+4.43 [−2.12, +10.61]	+1.27 [−0.95, +3.50]
Albania (Upper Mid.)	13.8% (+13.1 pp)	0.67% (€0.29B)	+0.26 [−0.36, +0.83]	€+0.11 [−0.16, +0.36]	+1.66 [−0.50, +3.86]
El Salvador (Upper Mid.)	11.2% (+15.6 pp)	0.44% (€0.28B)	+0.45 [−0.08, +0.99]	€+0.28 [−0.05, +0.62]	+1.13 [−1.04, +3.24]
Georgia (Upper Mid.)	10.7% (+16.1 pp)	0.55% (€0.40B)	+0.25 [−0.35, +0.80]	€+0.18 [−0.25, +0.57]	+1.60 [−0.55, +3.72]
Peru (Upper Mid.)	10.6% (+16.2 pp)	0.96% (€4.47B)	+0.43 [−0.09, +0.97]	€+2.02 [−0.41, +4.53]	+1.14 [−0.94, +3.20]
Mauritius (Upper Mid.)	10.4% (+16.4 pp)	0.48% (€0.14B)	+0.28 [−0.33, +0.89]	€+0.08 [−0.10, +0.26]	+1.95 [−0.33, +4.52]
Armenia (Upper Mid.)	9.8% (+17.0 pp)	0.56% (€0.27B)	+0.27 [−0.35, +0.85]	€+0.13 [−0.17, +0.41]	+1.91 [−0.35, +4.41]
Kazakhstan (Upper Mid.)	9.3% (+17.5 pp)	0.62% (€3.84B)	+0.25 [−0.37, +0.79]	€+1.55 [−2.28, +4.95]	+1.62 [−0.54, +3.68]
Mongolia (Upper Mid.)	7.9% (+18.9 pp)	1.02% (€0.50B)	+0.40 [−0.17, +0.98]	€+0.20 [−0.08, +0.48]	+1.26 [−0.86, +3.36]
Tunisia (Lower Mid.)	54.7% (+21.8 pp)	0.59% (€0.82B)	+0.47 [−0.08, +1.04]	€+0.66 [−0.11, +1.46]	+1.11 [−1.08, +3.13]
Lesotho (Lower Mid.)	44.7% (+31.8 pp)	0.40% (€0.02B)	+0.55 [−0.04, +1.25]	€+0.03 [−0.00, +0.07]	+0.18 [−3.29, +3.08]
Honduras (Lower Mid.)	40.9% (+35.5 pp)	0.72% (€0.43B)	+0.45 [−0.08, +0.99]	€+0.27 [−0.05, +0.59]	+1.18 [−0.93, +3.32]
Bangladesh (Lower Mid.)	29.9% (+46.6 pp)	0.88% (€10.75B)	+0.48 [−0.07, +1.06]	€+5.81 [−0.90, +12.95]	+1.15 [−1.09, +3.39]
Nepal (Lower Mid.)	24.2% (+52.3 pp)	0.95% (€1.22B)	+0.48 [−0.08, +1.07]	€+0.62 [−0.10, +1.37]	+1.15 [−1.18, +3.40]
Kyrgyz Republic (Lower Mid.)	4.9% (+71.6 pp)	2.60% (€1.04B)	+0.44 [−0.13, +1.00]	€+0.18 [−0.05, +0.40]	+1.29 [−0.83, +3.52]
Mali (Low)	173.8% (+31.0 pp)	0.11% (€0.06B)	+0.51 [−0.08, +1.16]	€+0.31 [−0.05, +0.70]	+0.56 [−3.11, +3.17]
Rwanda (Low)	102.8% (+102.0 pp)	0.67% (€0.25B)	+0.52 [−0.07, +1.19]	€+0.19 [−0.03, +0.45]	+0.38 [−3.32, +3.08]
Madagascar (Low)	99.1% (+105.7 pp)	0.34% (€0.15B)	+0.59 [−0.06, +1.34]	€+0.26 [−0.03, +0.60]	−0.24 [−3.97, +2.85]

C Propensity Overlap, Residual Covariance, and MCMC Diagnostics

Table 8 summarises covariate balance across all 19 pre-treatment covariates before and after inverse-propensity weighting (full table in `outputs/tables/tableA3_propensity_overlap_and_balance.csv`). Before weighting, treated country-waves (those spending above their income-group median per student) have higher total education spending as a share of GDP (SMD = +1.006), higher baseline non-resource tax-to-GDP (18.70% vs 15.70%, SMD = +0.465), and slower lagged GDP per person growth (SMD = −0.629) than control country-waves. After inverse-propensity weighting using $\hat{\pi}(X_{i,t})$, the mean absolute standardised mean difference across all 19 baseline covariates drops by more than half from 0.238 (maximum 1.006) to 0.105 (maximum 0.260), with 82.0% of observations lying inside the strict common-support region $\hat{\pi} \in [0.041, 0.960]$. This confirms that treated and control country-waves share substantial overlap across baseline economic and fiscal characteristics and that including $\hat{\pi}(X_{i,t})$ in the prognostic forest effectively balances pre-treatment confounders.

Table 9 reports the posterior mean residual standard deviations and all six cross-outcome residual

Table 8: Pre-treatment propensity overlap and standardised mean difference (SMD) covariate balance across key baseline predictors and the 19-covariate summary (+10-year sample, $N = 283$; strict common support spans $\hat{\pi} \in [0.041, 0.960]$ with 82.0% of observations inside the overlap region).

Baseline Pre-Treatment Covariate	Treated Mean	Control Mean	Unweighted SMD	IPW-Weighted SMD
Total Education Spend (% GDP)	5.32	3.87	+1.006	−0.089
Lagged GDP per Person Growth (%)	1.50	3.41	−0.629	−0.221
Baseline Non-Resource Tax / GDP (%)	18.70	15.70	+0.465	−0.068
Gross Tertiary Enrolment (%)	37.06	43.94	−0.286	−0.002
Baseline Direct Income Tax Share (%)	36.04	33.69	+0.165	−0.032
Log Real GDP per Person (PPP)	9.78	9.70	+0.084	−0.012
All 19 Covariates (Mean / Max |SMD|)	–	–	**0.238 / 1.006**	**0.105 / 0.260**

correlations from the MV-BCF model. These posterior correlations show why modelling all four tax outcomes jointly in a Multivariate BCF is preferable to fitting four independent regressions. Unobserved fiscal shocks to total non-resource tax/GDP and direct income tax/GDP are strongly positively correlated ($r = +0.60$ at +5 years and +0.62 at +10 years), as are shocks to direct income tax/GDP and the direct income tax share ($r = +0.71$ and +0.75), whereas shocks to the direct income tax share and the goods and services tax share are strongly negatively correlated ($r = -0.42$ at +5 years and −0.58 at +10 years) because the two revenue shares compete within the same national budget. Estimating the full 4×4 covariance matrix Σ via the Inverse-Wishart Gibbs step allows the tree updates to borrow strength across these strongly correlated fiscal innovations.

Table 9: Posterior mean cross-outcome residual correlations (with 95% CIs) and residual standard deviations from the four-outcome Multivariate Bayesian Causal Forest.

Residual Parameter Pair (MV-BCF)	+5y Mean [95% CI]	+10y Mean [95% CI]
1. Non-Res. Tax / GDP ↔ Income Tax / GDP	+0.60 [+0.51, +0.67]	+0.62 [+0.53, +0.71]
2. Income Tax / GDP ↔ Income Tax Share	+0.71 [+0.65, +0.77]	+0.75 [+0.68, +0.82]
3. Income Tax Share ↔ Goods Tax Share	−0.42 [−0.52, −0.32]	−0.58 [−0.69, −0.47]
4. Income Tax / GDP ↔ Goods Tax Share	−0.25 [−0.35, −0.13]	−0.25 [−0.38, −0.11]
5. Non-Res. Tax / GDP ↔ Goods Tax Share	+0.14 [+0.02, +0.26]	+0.35 [+0.20, +0.48]
6. Non-Res. Tax / GDP ↔ Income Tax Share	−0.04 [−0.15, +0.08]	+0.05 [−0.10, +0.21]
7. Residual SDs ($\sigma_1, \sigma_2, \sigma_3, \sigma_4$)	1.86, 1.01, 4.08, 4.23 pp	2.59, 1.38, 5.00, 4.98 pp

Table 10 and Figure 7 present the post-burn-in MCMC convergence diagnostics and posterior predictive check (PPC) fit metrics across the +5-year and +10-year MV-BCF models (1,250 retained draws). Across all eight horizon-by-outcome parameters in Table 10, split-chain $\hat{R}$ values lie between 1.000 and 1.020 (well below the standard 1.05 convergence threshold), effective sample sizes (ESS) range from 135 to 267, and Monte Carlo standard errors (MCSE) are small (0.008 to 0.069 pp, less than 9% of the corresponding posterior standard deviations). Furthermore, in-sample posterior predictive check R^2 values range from 0.192 to 0.592 across the macro-panel difference outcomes, and empirical 95% posterior predictive interval coverage (`Cov95`) lies between 95.4% and 97.5%, closely matching the nominal 95% target. Together with the stationary, well-mixed trace plots in Figure 7, these diagnostics confirm that our C++ Gibbs sampler converges reliably and is well calibrated to the observed distribution of fiscal changes.

Table 10: Post-burn-in MCMC convergence diagnostics (split-chain $\hat{R}$, effective sample size ESS, Monte Carlo standard error MCSE, and Geweke z-score) and in-sample posterior predictive check (PPC) R^2 and 95% CI coverage across the +5-year and +10-year MV-BCF models.

Model & Outcome	Horizon (N)	Mean (SD)	$\hat{R}$	ESS	MCSE	Geweke	PPC R^2	Cov95
1. MV-BCF Tax / GDP	+5y (372)	+0.020 (0.239)	**1.001**	265	0.015	−1.15	0.338	96.0%
2. MV-BCF Inc. Tax / GDP	+5y (372)	+0.210 (0.131)	**1.000**	267	0.008	−0.62	0.192	96.5%
3. MV-BCF Inc. Tax Share	+5y (372)	+1.241 (0.551)	**1.002**	223	0.037	−3.03	0.293	95.4%
4. MV-BCF Goods Share	+5y (372)	−0.287 (0.544)	**1.005**	182	0.040	+0.37	0.408	97.3%
5. MV-BCF Tax / GDP	+10y (283)	+0.446 (0.409)	**1.002**	202	0.029	−1.29	0.327	95.8%
6. MV-BCF Inc. Tax / GDP	+10y (283)	+0.344 (0.216)	**1.015**	198	0.015	+1.86	0.270	96.5%
7. MV-BCF Inc. Tax Share	+10y (283)	+1.479 (0.793)	**1.003**	198	0.056	+0.99	0.453	97.2%
8. MV-BCF Goods Share	+10y (283)	−0.559 (0.805)	**1.020**	135	0.069	−0.56	0.592	97.5%

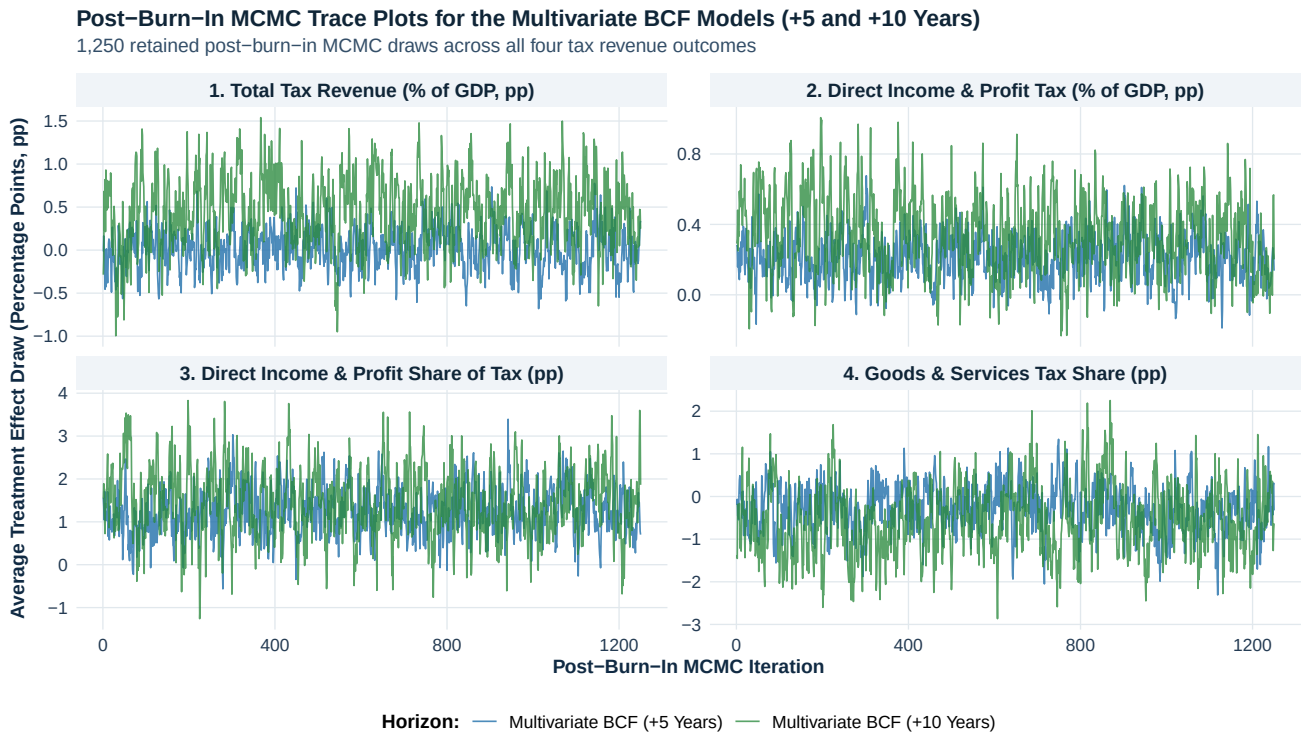


Figure 7: Post-burn-in MCMC trace plots across all four tax outcomes from the +5y and +10y Multivariate Bayesian Causal Forest (MV-BCF) sampler.

D Computational Environment and Energy Report

All data processing, C++ (`RcppArmadillo`) Gibbs sampling, and figure generation were executed on a single Apple Silicon workstation (`macOS`, R version 4.6.1, Apple Clang C++17 compiler). Table 11 reports both the local computational footprint of our statistical modelling pipeline (assuming an average 25 W CPU package power draw during single-core C++ tree sampling) and the cloud inference footprint of the large language model (LLM) queries used to assist with building the data pipeline, writing the C++ sampler, conducting multi-agent and external referee reviews, developing the companion web application, and editing the manuscript (evaluated at the Irish grid and European cloud average carbon intensity of 250 g CO_2e/kWh). Because our reversible-jump tree search and conjugate 4×4 Inverse-Wishart covariance updates are written directly in C++, fitting all 31 main, benchmark, and sensitivity models completes in just over 10 minutes (611.4 seconds) and consumes only 4.25 Wh of electricity (1.06 g CO_2e). By contrast, auditing our complete interaction logs across all five project stages reveals 753 LLM queries (comprising 1.61 million input and context tokens and 0.94 million generated output, reasoning, and tool-call tokens across the primary session, eight specialist reviewer and user-persona subagents, and two external LLM referee rounds). Applying published frontier LLM inference energy benchmarks (Luccioni et al. 2024; 0.30 Wh base overhead per agentic query plus 0.40 Wh per $1,000$ generated output tokens and 0.05 Wh per $1,000$ prefill input tokens), the cloud LLM queries consumed an estimated 682.20 Wh (170.55 g CO_2e), which is roughly 160 times the electrical energy required to run all 31 Bayesian MCMC models locally, bringing the combined project footprint to 686.45 Wh (171.61 g CO_2e, equivalent to driving a typical petrol passenger car approximately one kilometre).

Table 11: Computational wall-clock time, token volume, electrical energy use, and estimated carbon footprint across both the local C++ MCMC modelling pipeline and the cloud LLM queries used to help build, review, and write the paper.

Project Stage	Models Fitted / LLM Query & Token Volume	Time (s)	Energy (Wh)	CO_2e (g)
A. Local C++ MCMC Pipeline	31 four-outcome MV-BCF & benchmark fits (2,000 iter)	611.4	4.25	1.06
B1. LLM: Scoping & Data Pipeline	45 queries (75k input / 53k output tokens)	–	38.60	9.65
B2. LLM: C++ Sampler & Diagnostics	26 queries (37k input / 29k output tokens)	–	21.40	5.35
B3. LLM: Manuscript & Revisions	287 queries (477k input / 292k output tokens)	–	226.80	56.70
B4. LLM: Multi-Agent & Peer Review	261 queries (682k input / 361k output tokens)	–	256.80	64.20
B5. LLM: Policy Web App & Testing	134 queries (338k input / 204k output tokens)	–	138.60	34.65
Subtotal: Cloud LLM Assistance	**753 queries (1.61M input / 0.94M output tokens)**	–	**682.20**	**170.55**
Total Project Footprint (A + B)	**31 C++ MCMC fits + 753 LLM queries (2.55M tokens)**	**611.4**	**686.45**	**171.61**

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